GenMol / UDLM study through V13

34 configurations; 68 scheduled seed runs; 5,504 planned requests; 5,504 independently rescored accepted requests. Status: complete. No superiority or formal promotion is established.

V13 tests temperature placement using the same frozen CE checkpoint within each empirical/MASK-rich prior pair. Both modes use T=0.5, 128 predictor network evaluations per molecule, top-p=1, no Gibbs, and seeds 2100/2101 with 100 requests each. Raw mode converts CE to LOO before tempering; clean mode tempers CE logits first, converts to LOO, and uses bridge temperature one.

Four V13 settings

Configuration / accepted	Scoring	Quality % ± SD	Diversity ± SD
e_ce_raw_t050 200/200; {'completed': 2}	repaired	46.500 +/- 2.121	0.893 +/- 0.005
e_ce_raw_t050 200/200; {'completed': 2}	strict	34.500 +/- 2.121	0.878 +/- 0.002
e_ce_clean_t050 200/200; {'completed': 2}	repaired	52.000 +/- 9.899	0.891 +/- 0.001
e_ce_clean_t050 200/200; {'completed': 2}	strict	34.500 +/- 6.364	0.876 +/- 0.002
mask_ce_raw_t050 200/200; {'completed': 2}	repaired	49.500 +/- 3.536	0.882 +/- 0.001
mask_ce_raw_t050 200/200; {'completed': 2}	strict	42.000 +/- 2.828	0.873 +/- 0.003
mask_ce_clean_t050 200/200; {'completed': 2}	repaired	47.500 +/- 2.121	0.892 +/- 0.009
mask_ce_clean_t050 200/200; {'completed': 2}	strict	32.500 +/- 0.707	0.878 +/- 0.005

Both declared contrasts: clean temperature minus raw-LOO temperature

Entries below are signed within-seed differences followed by their equal-seed mean and sample SD. Quality is in percentage points; diversity is in raw units. Primary is empirical CE; secondary is MASK-rich CE. Pairing is by seed, not matched molecular trajectories.

Contrast / scoring	Metric	Seed 2100; 2101	Mean ± sample SD
primary / repaired	quality	+0.000; +11.000	+5.500 +/- 7.778
primary / repaired	diversity	-0.007; +0.003	-0.002 +/- 0.006
primary / strict	quality	-3.000; +3.000	+0.000 +/- 4.243
primary / strict	diversity	-0.005; +0.002	-0.002 +/- 0.005
secondary / repaired	quality	-6.000; +2.000	-2.000 +/- 5.657
secondary / repaired	diversity	+0.017; +0.003	+0.010 +/- 0.010
secondary / strict	quality	-12.000; -7.000	-9.500 +/- 3.536
secondary / strict	diversity	+0.007; +0.004	+0.006 +/- 0.001

Definitions and limits

Quality = within-seed unique valid molecules with QED >= 0.6 and SA <= 4 / all requested samples. QED measures drug-likeness (higher favored); SA estimates synthetic accessibility (lower favored). Validity = valid/requested; uniqueness = unique valid/valid. Diversity uses the released PyTDC metric on unique valid molecules. Repaired scoring applies SAFE repair and keeps the largest component; strict SAFE disables repair. Both remove tokenizer special tokens before decoding, so exact final control-token counts are recorded separately in overview.json.

Every ± value is sample SD across two seeds, not a confidence interval. Missing or failed seeds withhold configuration and paired means/SD; retained partial source aggregates are explicitly separate. All four validity/uniqueness/quality/diversity outcomes and pair differences remain in the untouched V13 appendix and JSON/CSV.

These are adaptive engineering pilots across 34 settings, not independent confirmatory tests. V5/V6 use 128 requests per setting; V9/V10/V12/V13 use 200. The local MDLM reference uses 3 × 1,000 requests, repaired quality 85.8%; strict quality is not directly comparable to that repaired reference. The baseline's 3,000 requests and failed V4 diagnostic's 32 requests are outside UDLM accepted totals. Different study seeds cannot support paired cross-study effects. Final UDLM seeds 0/1/2 remain reserved.

V13 adds no training. Historical R/S/E adaptation used 1,000 updates × batch16 = 16,000 exposures. Empirical CE and MASK-rich CE each used a fresh MDLM 50k EMA initialization, 1,000 updates × batch128 = 128,000 exposures, seed1500; the changed prior changes denoising difficulty. V8 CT's failed post-exit controller and separate checkpoint audit, unstarted original CE arm, separate successful V8b CE, V11 prelaunch failure with no trained checkpoint, and successful V11b MASK CE remain preserved. Inference costs are not uniform across all studies: V10 includes 512 NFE, four times 128. V13's planned molecule-NFE budget is 102,400.

Appendix order: untouched terminal V13 report, then the preserved 78-page overview through V12 (including its original reports, training/configuration details, prior/checkpoint identities, failures, and historical caveats). Older appendices retain the statements current when they were written. All original bytes, source relocations, inputs, and output hashes are recorded in the new manifest.

GenMol UDLM engineering exploration

engineering-v13-ce-temperature-space | complete | 2026-09-07T13:15:46.418543+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

Each temperature-space pair uses the same frozen CE checkpoint and prior. Clean-denoiser temperature is applied before conversion; raw-LOO temperature is applied after conversion. This inference ablation does not add training or backbone evaluations.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED >= 0.6 and SA <= 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Predeclared contrasts subtract raw-LOO-temperature from clean-denoiser-temperature within the same seed and decoding branch. Every declared contrast is retained, including negative outcomes. Paired mean and sample SD are withheld unless all declared seed pairs define that metric; unavailable pairs remain visible. Sample SD describes seed spread, not a confidence interval. Pairing seed labels does not imply coupled molecular trajectories.

Prospective sampling design

{"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "parameterization_controls": "Both arms use the same CE checkpoint within each prior pair. Control applies temperature after CE-to-LOO conversion; treatment tempers clean logits before conversion and fixes reverse bridge temperature to one. NFE is unchanged.", "predictor_transitions_per_molecule": 128, "randomness": 0.0, "raw_loo_top_p": 1.0, "temperature": 0.5, "temperature_space_comparison": {"primary": {"control_config": "e_ce_raw_t050", "temperature": 0.5, "treatment_config": "e_ce_clean_t050"}, "secondary": {"control_config": "mask_ce_raw_t050", "temperature": 0.5, "treatment_config": "mask_ce_clean_t050"}}, "temperature_space_semantics": {"raw_loo": "loo_conversion_then_temperature_then_reverse_bridge", "x0_denoiser": "clean_denoiser_temperature_then_loo_conversion_then_reverse_bridge_at_one"}, "total_requested_molecule_network_evaluations": 102400}

The prospective design was fixed while V12 was running, before reading any MASK-rich molecular outcome. It discloses incidental viewing of one unfinished empirical T1 log. No V12 outcome selected a prior, temperature or excluded arm.

Temperature 0.5 was informed by earlier V5/V6/V9/V10 engineering studies. Both empirical and MASK-rich CE checkpoints are retained; this is a new inference hypothesis, not a bug fix or additional training.

Within each pair only temperature_space changes: same frozen checkpoint, prior, EMA, temperature 0.5, 128 predictor NFE, top-p 1, no Gibbs and other sampling controls. Clean temperature acts before conversion; its reverse bridge temperature is one.

Two seeds of 100 requests are small. Pair by seed, not molecular trajectory. Report every configuration and failure, both strict/repaired branches, all four metrics and signed clean-temperature-minus-raw-LOO-temperature contrasts, including negative differences.

Quality counts within-seed unique valid molecules with QED >= 0.6 and SA <= 4 divided by all requests. Withhold each paired mean/SD unless both declared seed pairs define the metric; SD is descriptive, not a confidence interval.

Disclose final editable MASK and other control counts with the editable-position denominator from sampled_token_control_audit. Decoded strings skip specials and cannot supply these counts; report recovery and component-selection frequencies with denominators.

Training/checkpoint references are inherited unchanged from the accepted V12 protocol; no molecular metrics are used by this preparation. Original V11 failed before training and V8 CT remains a failed controller with a separate checkpoint audit; V11b and V8b are distinct successful follow-ups.

Each CE checkpoint has 1,000 adaptation updates and 128,000 configured exposures from MDLM 50k EMA. These are not necessarily distinct molecules, equal exposures do not imply equal runtime, and final finiteness does not establish every update was finite.

Local MDLM quality 0.858 and paper GenMol V1 quality 0.846 use different training/sample budgets and remain contextual. No superiority, automatic promotion or final-seed evaluation; seeds 0/1/2 remain reserved.

Temperature-space comparison

Both arms use the same frozen clean CE checkpoint within each pair. The control tempers raw LOO logits after denoiser conversion. The treatment tempers clean logits before conversion and uses bridge temperature one. The chosen temperature, prior and all other sampling controls are held fixed within each pair. Neither rule adds a backbone evaluation; this is a new inference hypothesis, not additional training.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"E_CE": {"checkpoint_sha256": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "checkpoint_size_bytes": 1636492999, "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020}, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b", "size_bytes": 3161}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size_bytes": 8621}, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}, "udlm_prior_metadata": {"active_token_ids_sha256": "2fd62050919fdf6ee578df881733e47834f662406c219a1f247116e04396a5ae", "active_vocab_size": 1880, "antithetic_sampling": true, "comparison_role": "empirical_prior_treatment", "excluded_token_ids": [], "frequency_active_token_count": 517090, "frequency_artifact_path": "experiments/udlm/token_frequency/train_first_10000.json", "frequency_artifact_schema_version": 1, "frequency_artifact_sha256": 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1880, "mask_mixture_weight": 0.9, "mask_token_id": 4, "noise_eps": 0.001, "objective_scope": "model_dependent_ct_integrand_without_parameter_independent_endpoint_kl", "prior_source": "mask_point_mass_mixture_with_pinned_smoothed_frequency", "process_family": "rank_one_continuous_categorical", "sampling_eps": 0.001, "schedule_variant": "schedule_consistent_residual_forward_and_loss", "schema_version": 1, "stationary_probs_sha256": "07d011c07bdac8fdbe096f514ded6e2d1b6850268c48bec8dd0e3f3853d013eb", "tokenizer_json_sha256": "0db5f4dbdc7e8ff759e98483759611a426e187ee7f3f0a91edc8800abe7bf140", "tokenizer_repo_id": "datamol-io/safe-gpt", "tokenizer_revision": "3d5fa0988383e898d5ac5db7cd52bf715bc37061", "uniform_mixture_weight": 0.0002, "variant": "mask_rich_empirical"}}}, "common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_definition": "Empirical-arm configuration after removing parameterization and normalizing callback.dirpath. The MASK arm matches after restoring empirical_frequency and removing mask_mixture_weight.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846}, "original_v11_failure": {"relative_path": "output/udlm/engineering_v11/mask_ce_1000_b128_w2/terminal_manifest.json", "sha256": "6a4aa47e1f7b76ad5efc3ce03cd3e4a55c0db4d95778b0cf60cc55cab7207408", "size_bytes": 14510}, "original_v8_ct_context": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166}, "resolved_config_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}, "original_v8_protocol": {"relative_path":

"experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak_learning_rate": 0.0003, "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50}, "training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdlm_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
e_ce_raw_t050	2100	128	128	0
e_ce_raw_t050	2101	128	128	0
e_ce_clean_t050	2100	128	128	0
e_ce_clean_t050	2101	128	128	0
mask_ce_raw_t050	2100	128	128	0
mask_ce_raw_t050	2101	128	128	0
mask_ce_clean_t050	2100	128	128	0
mask_ce_clean_t050	2101	128	128	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
e_ce_raw_t050	2100	clean CE	x0_denoiser	ema
e_ce_raw_t050	2101	clean CE	x0_denoiser	ema
e_ce_clean_t050	2100	clean CE	x0_denoiser	ema

Config	Seed	Objective	Parameterization	Weights
e_ce_clean_t050	2101	clean CE	x0_denoiser	ema
mask_ce_raw_t050	2100	clean CE	x0_denoiser	ema
mask_ce_raw_t050	2101	clean CE	x0_denoiser	ema
mask_ce_clean_t050	2100	clean CE	x0_denoiser	ema
mask_ce_clean_t050	2101	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_ce_raw_t050	2/2	0.9900 +/- 0.0141	0.9798 +/- 0.0003	0.4650 +/- 0.0212	0.8934 +/- 0.0053
e_ce_clean_t050	2/2	0.9750 +/- 0.0212	1.0000 +/- 0.0000	0.5200 +/- 0.0990	0.8914 +/- 0.0011
mask_ce_raw_t050	2/2	0.9900 +/- 0.0000	0.9949 +/- 0.0071	0.4950 +/- 0.0354	0.8817 +/- 0.0007
mask_ce_clean_t050	2/2	0.9850 +/- 0.0071	0.9949 +/- 0.0072	0.4750 +/- 0.0212	0.8916 +/- 0.0091

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_ce_raw_t050	2/2	0.6650 +/- 0.0495	1.0000 +/- 0.0000	0.3450 +/- 0.0212	0.8777 +/- 0.0022
e_ce_clean_t050	2/2	0.6400 +/- 0.0566	1.0000 +/- 0.0000	0.3450 +/- 0.0636	0.8761 +/- 0.0024
mask_ce_raw_t050	2/2	0.8050 +/- 0.0212	1.0000 +/- 0.0000	0.4200 +/- 0.0283	0.8728 +/- 0.0034
mask_ce_clean_t050	2/2	0.6650 +/- 0.0495	1.0000 +/- 0.0000	0.3250 +/- 0.0071	0.8783 +/- 0.0048

Predeclared paired clean-temperature minus raw-LOO-temperature contrasts

Differences use the same seed and decoding branch, with equal seed weights. Values are on the metric's 0-to-1 scale: 0.01 is one percentage point for quality. All declared pairs must define a metric before its mean and sample SD are shown. SD is seed spread, not a confidence interval. Seed labels do not imply paired molecules.

primary | temperature 0.5: e_ce_clean_t050 minus e_ce_raw_t050

released_comparable

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
validity	2100	+0.980000	+0.960000	-0.020000	completed
validity	2101	+1.000000	+0.990000	-0.010000	completed
uniqueness	2100	+0.979592	+1.000000	+0.020408	completed
uniqueness	2101	+0.980000	+1.000000	+0.020000	completed
quality	2100	+0.450000	+0.450000	+0.000000	completed
quality	2101	+0.480000	+0.590000	+0.110000	completed
diversity	2100	+0.897172	+0.890625	-0.006547	completed
diversity	2101	+0.889607	+0.892243	+0.002636	completed

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
validity	2/2	-0.015000	0.007071
uniqueness	2/2	+0.020204	0.000289
quality	2/2	+0.055000	0.077782
diversity	2/2	-0.001956	0.006493

strict

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
validity	2100	+0.630000	+0.600000	-0.030000	completed
validity	2101	+0.700000	+0.680000	-0.020000	completed
uniqueness	2100	+1.000000	+1.000000	+0.000000	completed
uniqueness	2101	+1.000000	+1.000000	+0.000000	completed
quality	2100	+0.330000	+0.300000	-0.030000	completed
quality	2101	+0.360000	+0.390000	+0.030000	completed

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
diversity	2100	+0.879281	+0.874420	-0.004861	completed
diversity	2101	+0.876110	+0.877813	+0.001703	completed

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
validity	2/2	-0.025000	0.007071
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.000000	0.042426
diversity	2/2	-0.001579	0.004641

secondary | temperature 0.5: mask_ce_clean_t050 minus mask_ce_raw_t050

released_comparable

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
validity	2100	+0.990000	+0.980000	-0.010000	completed
validity	2101	+0.990000	+0.990000	+0.000000	completed
uniqueness	2100	+1.000000	+0.989796	-0.010204	completed
uniqueness	2101	+0.989899	+1.000000	+0.010101	completed
quality	2100	+0.520000	+0.460000	-0.060000	completed
quality	2101	+0.470000	+0.490000	+0.020000	completed
diversity	2100	+0.881186	+0.898097	+0.016911	completed
diversity	2101	+0.882173	+0.885188	+0.003015	completed

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
validity	2/2	-0.005000	0.007071
uniqueness	2/2	-0.000052	0.014358
quality	2/2	-0.020000	0.056569

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
diversity	2/2	+0.009963	0.009826

strict

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
validity	2100	+0.820000	+0.630000	-0.190000	completed
validity	2101	+0.790000	+0.700000	-0.090000	completed
uniqueness	2100	+1.000000	+1.000000	+0.000000	completed
uniqueness	2101	+1.000000	+1.000000	+0.000000	completed
quality	2100	+0.440000	+0.320000	-0.120000	completed
quality	2101	+0.400000	+0.330000	-0.070000	completed
diversity	2100	+0.875190	+0.881751	+0.006561	completed
diversity	2101	+0.870420	+0.874920	+0.004500	completed

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
validity	2/2	-0.140000	0.070711
uniqueness	2/2	+0.000000	0.000000
quality	2/2	-0.095000	0.035355
diversity	2/2	+0.005531	0.001458

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_ce_raw_t050	2100	completed	100	100	0.4500	0.3300	17.69
e_ce_raw_t050	2101	completed	100	100	0.4800	0.3600	18.02
e_ce_clean_t050	2100	completed	100	100	0.4500	0.3000	19.79
e_ce_clean_t050	2101	completed	100	100	0.5900	0.3900	20.84

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
mask_ce_raw_t050	2100	completed	100	100	0.5200	0.4400	14.10
mask_ce_raw_t050	2101	completed	100	100	0.4700	0.4000	14.95
mask_ce_clean_t050	2100	completed	100	100	0.4600	0.3200	14.24
mask_ce_clean_t050	2101	completed	100	100	0.4900	0.3300	16.00

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: daf8c8108772e0ebcbdd39f15dab07478fe8945ba6b33f4b68858a8536ad7319

v13-e-ce-raw-t050 / seed 2100: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison":

"strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-e-ce-raw-t050/seed_2100/raw_samples.csv", "sha256": "f625f31422a820d1b9384b9b3fc38668b42a860f293132ae241de2233509362d", "size_bytes": 32295}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-e-ce-raw-t050 / seed 2101: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-e-ce-raw-t050/seed_2101/raw_samples.csv", "sha256": "c235b82d57e6f54bdc28b304ec1f6a7c569dcf8b240a0694df52bd7c5727a70e", "size_bytes": 33222}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-e-ce-clean-t050 / seed 2100: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "temperature_space": "x0_denoiser"}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "reverse_bridge_temperature": 1.0, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "temperature_application": "clean_denoiser_before_loo_conversion", "temperature_space": "x0_denoiser"}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-e-ce-clean-t050/seed_2100/raw_samples.csv", "sha256": "598044d70dfcef4bc968f25dbfa7537f407c31806fa64c8003f72950ee3504a5", "size_bytes": 32059}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-e-ce-clean-t050 / seed 2101: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "temperature_space": "x0_denoiser"}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "reverse_bridge_temperature": 1.0, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "temperature_application": "clean_denoiser_before_loo_conversion", "temperature_space": "x0_denoiser"}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-e-ce-clean-t050/seed_2101/raw_samples.csv", "sha256": "66b3a97cdbaa2571bb4b6fb3643d42b63f58d922d74947c40609e9e20f69e210", "size_bytes": 33251}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-mask-ce-raw-t050 / seed 2100: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid":

"GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-mask-ce-raw-t050/seed_2100/raw_samples.csv", "sha256": "1eb3b11226c43e83db59ecc80c5bacc10051629d88f44c565b172e8d3dfbade9", "size_bytes": 34211}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-mask-ce-raw-t050 / seed 2101: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-mask-ce-raw-t050/seed_2101/raw_samples.csv", "sha256": "a872bdcac13280e7f805594abe6eafae6afe1d3b3f4289e0e82035bf60e8517d", "size_bytes": 34038}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-mask-ce-clean-t050 / seed 2100: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "temperature_space": "x0_denoiser"}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "reverse_bridge_temperature": 1.0, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "temperature_application": "clean_denoiser_before_loo_conversion", "temperature_space": "x0_denoiser"}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-mask-ce-clean-t050/seed_2100/raw_samples.csv", "sha256": "557a800a61fde29b1f4588a35aa923bd7090906aa886e5f8d3c5d2938775aef7", "size_bytes": 32216}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-mask-ce-clean-t050 / seed 2101: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "temperature_space": "x0_denoiser"}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "reverse_bridge_temperature": 1.0, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "temperature_application": "clean_denoiser_before_loo_conversion", "temperature_space": "x0_denoiser"}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-mask-ce-clean-t050/seed_2101/raw_samples.csv", "sha256": "763d698ec814de379f24772cd6c87d3323d921a0cee54f588ad7a711d3beb39c", "size_bytes": 33316}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

GenMol / UDLM engineering study through V12

30 sampling configurations; 60 scheduled runs; 4704 independently rescored accepted requests out of 4704 planned. Status: complete. These are sampling configurations, not independent trained models.

No superiority established. All engineering settings use two seeds: V5/V6 request 64 per seed, V9/V10/V12 request 100 per seed. The local MDLM reference uses 3 x 1,000; the paper reports 3 runs of 1,000. These different training and sampling budgets are context, not a matched superiority test. Final UDLM seeds 0/1/2 remain reserved.

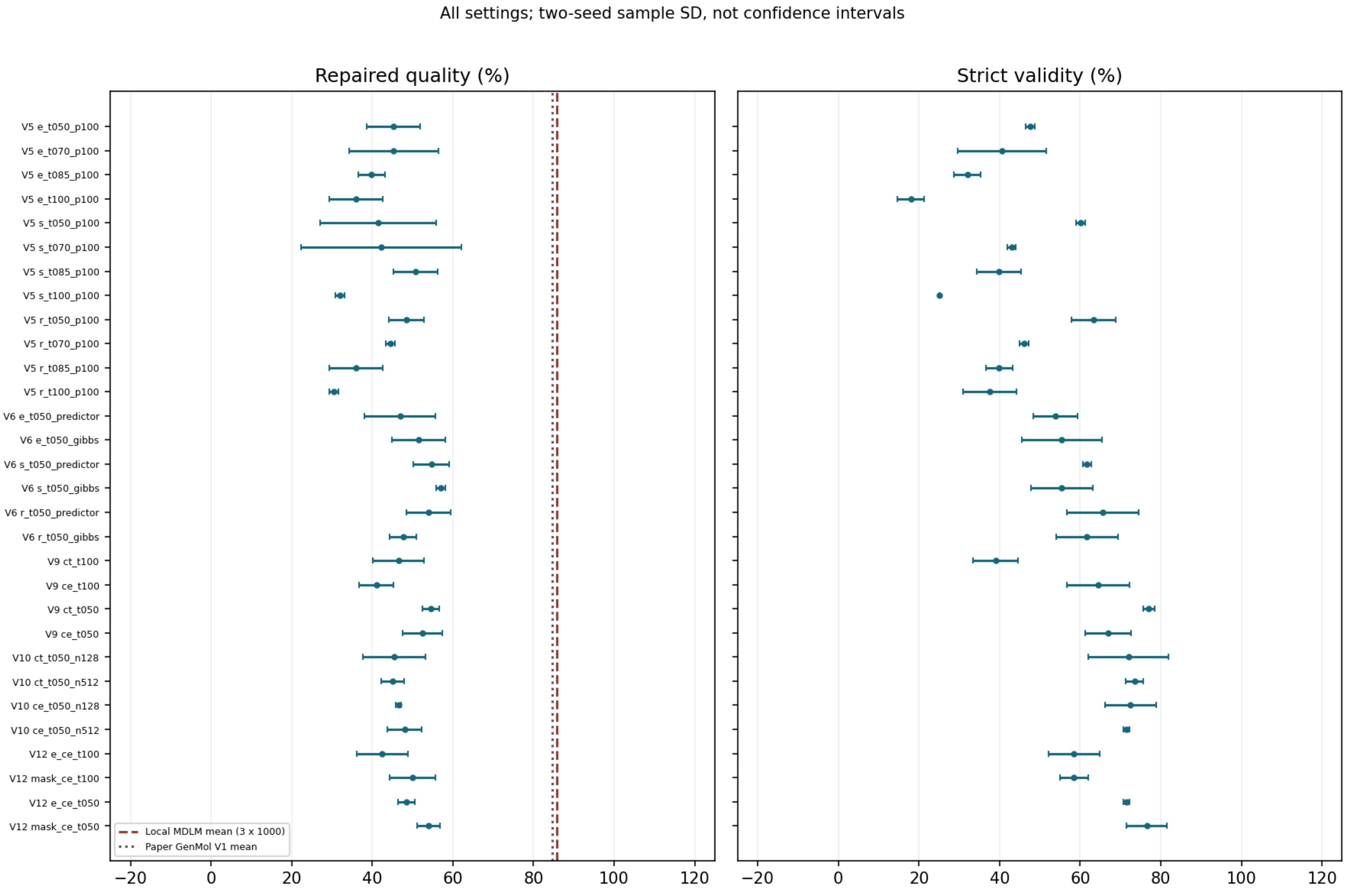
Study	Configs	Scheduled runs	Accepted / planned requests	Run status counts
V5	12	24	1536 / 1536	{'completed': 24}
V6	6	12	768 / 768	{'completed': 12}
V9	4	8	800 / 800	{'completed': 8}
V10	4	8	800 / 800	{'completed': 8}
V12	4	8	800 / 800	{'completed': 8}

The separate failed V4 diagnostic requested 32 samples and remains outside these accepted totals. Original V8 CT controller failure and V11 prelaunch failure remain failures; their follow-ups and the separate CT checkpoint audit retain distinct evidence.

Quality counts within-seed unique valid molecules with QED >= 0.6 and SA <= 4 divided by all requested samples. QED measures drug-likeness (higher is favored); SA estimates synthetic accessibility (lower is favored). Validity uses requests; uniqueness uses valid samples; diversity is the released PyTDC metric on unique valid molecules. Strict SAFE decoding disables repair; released-compatible decoding repairs and selects the largest component. Both start after tokenizer special-token removal.

Every mean and +/- value below is an equal-seed mean and sample SD, not a confidence interval. Means are withheld unless both declared seeds completed and defined the metric. Settings and temperatures were selected adaptively across studies; changed-seed comparisons are not paired treatment effects. Preserved appendices retain their original historical statements and dates.

All configurations: quality and strict validity



V5: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
e_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Repaired	97.656 +/- 1.105	100.000 +/- 0.000	45.312 +/- 6.629	0.892 +/- 0.006
e_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Strict	47.656 +/- 1.105	100.000 +/- 0.000	22.656 +/- 5.524	0.878 +/- 0.002
e_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Repaired	98.438 +/- 0.000	99.206 +/- 1.122	45.312 +/- 11.049	0.899 +/- 0.003
e_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Strict	40.625 +/- 11.049	100.000 +/- 0.000	21.094 +/- 9.944	0.872 +/- 0.015
e_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Repaired	94.531 +/- 1.105	96.680 +/- 2.376	39.844 +/- 3.315	0.904 +/- 0.005
e_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Strict	32.031 +/- 3.315	100.000 +/- 0.000	13.281 +/- 3.315	0.883 +/- 0.010
e_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Repaired	93.750 +/- 4.419	99.138 +/- 1.219	35.938 +/- 6.629	0.920 +/- 0.012
e_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Strict	17.969 +/- 3.315	100.000 +/- 0.000	9.375 +/- 4.419	0.884 +/- 0.007
s_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Repaired	97.656 +/- 1.105	99.194 +/- 1.140	41.406 +/- 14.363	0.897 +/- 0.004
s_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Strict	60.156 +/- 1.105	100.000 +/- 0.000	23.438 +/- 8.839	0.878 +/- 0.004
s_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Repaired	97.656 +/- 1.105	100.000 +/- 0.000	42.188 +/- 19.887	0.898 +/- 0.006
s_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Strict	42.969 +/- 1.105	100.000 +/- 0.000	21.094 +/- 1.105	0.864 +/- 0.004
s_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Repaired	96.094 +/- 1.105	100.000 +/- 0.000	50.781 +/- 5.524	0.904 +/- 0.004
s_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Strict	39.844 +/- 5.524	100.000 +/- 0.000	24.219 +/- 3.315	0.871 +/- 0.004
s_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Repaired	92.969 +/- 7.734	98.413 +/- 2.245	32.031 +/- 1.105	0.924 +/- 0.005
s_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Strict	25.000 +/- 0.000	100.000 +/- 0.000	11.719 +/- 1.105	0.886 +/- 0.002
r_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Repaired	99.219 +/- 1.105	100.000 +/- 0.000	48.438 +/- 4.419	0.881 +/- 0.001
r_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Strict	63.281 +/- 5.524	100.000 +/- 0.000	29.688 +/- 4.419	0.869 +/- 0.001
r_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Repaired	97.656 +/- 3.315	100.000 +/- 0.000	44.531 +/- 1.105	0.894 +/- 0.002
r_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Strict	46.094 +/- 1.105	100.000 +/- 0.000	19.531 +/- 5.524	0.875 +/- 0.002
r_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Repaired	96.875 +/- 2.210	99.206 +/- 1.122	35.938 +/- 6.629	0.911 +/- 0.002
r_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Strict	39.844 +/- 3.315	100.000 +/- 0.000	21.094 +/- 5.524	0.877 +/- 0.010
r_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Repaired	93.750 +/- 0.000	99.167 +/- 1.179	30.469 +/- 1.105	0.911 +/- 0.005
r_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Strict	37.500 +/- 6.629	100.000 +/- 0.000	14.062 +/- 2.210	0.881 +/- 0.003

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

V6: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
e_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Repaired	96.875 +/- 0.000	99.194 +/- 1.140	46.875 +/- 8.839	0.896 +/- 0.010
e_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Strict	53.906 +/- 5.524	100.000 +/- 0.000	31.250 +/- 2.210	0.887 +/- 0.006
e_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Repaired	98.438 +/- 2.210	98.412 +/- 0.036	51.562 +/- 6.629	0.889 +/- 0.003
e_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Strict	55.469 +/- 9.944	100.000 +/- 0.000	29.688 +/- 0.000	0.874 +/- 0.003
s_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Repaired	98.438 +/- 2.210	99.194 +/- 1.140	54.688 +/- 4.419	0.882 +/- 0.006
s_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Strict	61.719 +/- 1.105	100.000 +/- 0.000	33.594 +/- 1.105	0.866 +/- 0.007
s_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Repaired	98.438 +/- 2.210	97.606 +/- 1.176	57.031 +/- 1.105	0.892 +/- 0.005
s_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Strict	55.469 +/- 7.734	100.000 +/- 0.000	36.719 +/- 1.105	0.878 +/- 0.004
r_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Repaired	100.000 +/- 0.000	98.438 +/- 2.210	53.906 +/- 5.524	0.885 +/- 0.007
r_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Strict	65.625 +/- 8.839	100.000 +/- 0.000	36.719 +/- 3.315	0.874 +/- 0.001
r_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Repaired	97.656 +/- 1.105	100.000 +/- 0.000	47.656 +/- 3.315	0.893 +/- 0.012
r_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Strict	61.719 +/- 7.734	100.000 +/- 0.000	32.031 +/- 5.524	0.877 +/- 0.008

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

V9: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
ct_t100 / 1.0 / 128P	200/200; [1600, 1601]	Repaired	97.500 +/- 2.121	99.479 +/- 0.737	46.500 +/- 6.364	0.909 +/- 0.000
ct_t100 / 1.0 / 128P	200/200; [1600, 1601]	Strict	39.000 +/- 5.657	100.000 +/- 0.000	21.000 +/- 1.414	0.886 +/- 0.007
ce_t100 / 1.0 / 128P	200/200; [1600, 1601]	Repaired	98.000 +/- 1.414	99.485 +/- 0.729	41.000 +/- 4.243	0.903 +/- 0.004
ce_t100 / 1.0 / 128P	200/200; [1600, 1601]	Strict	64.500 +/- 7.778	100.000 +/- 0.000	26.000 +/- 7.071	0.887 +/- 0.005
ct_t050 / 0.5 / 128P	200/200; [1600, 1601]	Repaired	98.500 +/- 0.707	99.495 +/- 0.714	54.500 +/- 2.121	0.884 +/- 0.003
ct_t050 / 0.5 / 128P	200/200; [1600, 1601]	Strict	77.000 +/- 1.414	100.000 +/- 0.000	44.000 +/- 1.414	0.868 +/- 0.008
ce_t050 / 0.5 / 128P	200/200; [1600, 1601]	Repaired	99.000 +/- 0.000	99.495 +/- 0.714	52.500 +/- 4.950	0.888 +/- 0.006
ce_t050 / 0.5 / 128P	200/200; [1600, 1601]	Strict	67.000 +/- 5.657	100.000 +/- 0.000	39.500 +/- 3.536	0.870 +/- 0.001

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

V10: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
ct_t050_n128 / 0.5 / 128P	200/200; [1700, 1701]	Repaired	100.000 +/- 0.000	100.000 +/- 0.000	45.500 +/- 7.778	0.892 +/- 0.006
ct_t050_n128 / 0.5 / 128P	200/200; [1700, 1701]	Strict	72.000 +/- 9.899	100.000 +/- 0.000	35.500 +/- 12.021	0.875 +/- 0.002
ct_t050_n512 / 0.5 / 512P	200/200; [1700, 1701]	Repaired	99.500 +/- 0.707	100.000 +/- 0.000	45.000 +/- 2.828	0.891 +/- 0.001
ct_t050_n512 / 0.5 / 512P	200/200; [1700, 1701]	Strict	73.500 +/- 2.121	100.000 +/- 0.000	32.500 +/- 3.536	0.876 +/- 0.005
ce_t050_n128 / 0.5 / 128P	200/200; [1700, 1701]	Repaired	99.000 +/- 1.414	99.490 +/- 0.722	46.500 +/- 0.707	0.893 +/- 0.003
ce_t050_n128 / 0.5 / 128P	200/200; [1700, 1701]	Strict	72.500 +/- 6.364	100.000 +/- 0.000	36.000 +/- 0.000	0.878 +/- 0.001
ce_t050_n512 / 0.5 / 512P	200/200; [1700, 1701]	Repaired	98.500 +/- 0.707	100.000 +/- 0.000	48.000 +/- 4.243	0.893 +/- 0.001
ce_t050_n512 / 0.5 / 512P	200/200; [1700, 1701]	Strict	71.500 +/- 0.707	100.000 +/- 0.000	38.500 +/- 3.536	0.878 +/- 0.005

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

V12: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
e_ce_t100 / 1.0 / 128P	200/200; [2000, 2001]	Repaired	99.000 +/- 1.414	99.500 +/- 0.707	42.500 +/- 6.364	0.901 +/- 0.003
e_ce_t100 / 1.0 / 128P	200/200; [2000, 2001]	Strict	58.500 +/- 6.364	100.000 +/- 0.000	27.500 +/- 2.121	0.885 +/- 0.006
mask_ce_t100 / 1.0 / 128P	200/200; [2000, 2001]	Repaired	99.500 +/- 0.707	99.500 +/- 0.707	50.000 +/- 5.657	0.905 +/- 0.000
mask_ce_t100 / 1.0 / 128P	200/200; [2000, 2001]	Strict	58.500 +/- 3.536	100.000 +/- 0.000	34.000 +/- 5.657	0.882 +/- 0.001
e_ce_t050 / 0.5 / 128P	200/200; [2000, 2001]	Repaired	98.000 +/- 1.414	100.000 +/- 0.000	48.500 +/- 2.121	0.894 +/- 0.007
e_ce_t050 / 0.5 / 128P	200/200; [2000, 2001]	Strict	71.500 +/- 0.707	100.000 +/- 0.000	37.500 +/- 3.536	0.883 +/- 0.008
mask_ce_t050 / 0.5 / 128P	200/200; [2000, 2001]	Repaired	99.500 +/- 0.707	99.495 +/- 0.714	54.000 +/- 2.828	0.886 +/- 0.004
mask_ce_t050 / 0.5 / 128P	200/200; [2000, 2001]	Strict	76.500 +/- 4.950	100.000 +/- 0.000	42.000 +/- 4.243	0.876 +/- 0.000

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

Training outcomes remain separate

Training event	Outcome	Updates / configured exposures	Evidence / interpretation
Historical E	Audited frozen training checkpoint	1000 / 16000	Batch16; seed17; checkpoint dce870e8d634
Historical R	Audited frozen training checkpoint	1000 / 16000	Batch16; seed17; checkpoint d0310d2e2402
Historical S	Audited frozen training checkpoint	1000 / 16000	Batch16; seed17; checkpoint 100f467b9476
V7 throughput pilot	completed	20 / 2560	Source e78267b09810; checkpoint 07ad66498cca; final finite tensors 1155
V8 CT controller	failed	not accepted by controller / None	Source c8434dda105c; separate post-exit audit accepted step1000, checkpoint 48986899c401; failed receipt unchanged
V8b empirical CE	completed	1000 / 128000	Source b84f21ba1f30; checkpoint b7d674ed5bdd; final finite tensors 1156
V11 MASK-rich CE	failed	not accepted by controller / None	Source 46ec3b0bc0c5; no child or checkpoint
V11b MASK-rich CE	completed	1000 / 128000	Source be18a3244a71; checkpoint 62299de99d8c; final finite tensors 1157
Original V8 campaign	failed	CT then campaign stopped	Original CE arm never launched; V8b is a separate fresh follow-up.

All continuations start from original MDLM 50k EMA with fresh optimizer/EMA. Historical R/S/E add 1,000 updates at batch 16 (16,000 exposures). V8 CT, V8b CE and V11b CE add 1,000 at batch 128 (128,000 exposures), eight times as many exposures. Exposures are not necessarily distinct molecules. Final checkpoint finiteness does not prove every update was finite. Original MDLM 50k training has a different budget.

V12: signed MASK-rich minus empirical CE contrasts

Contrast / T	Branch / metric	Seed2000 delta	Seed2001 delta	Mean +/- sample SD
primary / 1.0	released_comparable / validity	+0.000	+1.000	+0.500 +/- 0.707
primary / 1.0	released_comparable / uniqueness	+0.000	+0.000	+0.000 +/- 0.000
primary / 1.0	released_comparable / quality	+16.000	-1.000	+7.500 +/- 12.021
primary / 1.0	released_comparable / diversity	+0.006	+0.001	+0.004 +/- 0.004
primary / 1.0	strict / validity	-2.000	+2.000	+0.000 +/- 2.828
primary / 1.0	strict / uniqueness	+0.000	+0.000	+0.000 +/- 0.000
primary / 1.0	strict / quality	+12.000	+1.000	+6.500 +/- 7.778
primary / 1.0	strict / diversity	+0.001	-0.007	-0.003 +/- 0.005
secondary / 0.5	released_comparable / validity	+1.000	+2.000	+1.500 +/- 0.707
secondary / 0.5	released_comparable / uniqueness	+0.000	-1.010	-0.505 +/- 0.714
secondary / 0.5	released_comparable / quality	+6.000	+5.000	+5.500 +/- 0.707
secondary / 0.5	released_comparable / diversity	-0.006	-0.010	-0.008 +/- 0.003
secondary / 0.5	strict / validity	+8.000	+2.000	+5.000 +/- 4.243
secondary / 0.5	strict / uniqueness	+0.000	+0.000	+0.000 +/- 0.000
secondary / 0.5	strict / quality	+5.000	+4.000	+4.500 +/- 0.707
secondary / 0.5	strict / diversity	-0.001	-0.012	-0.007 +/- 0.008

Percentage-point differences for validity, uniqueness and quality; raw units for diversity. Pairing is by declared seed, not molecular trajectory. A failed seed withholds the paired mean/SD. Temperature 1 is primary; 0.5 is secondary and acts after CE-to-LOO conversion.

V12: final editable MASK occurrences

Configuration	Seed	Final MASK / editable positions	Rows with MASK / requests	All generated control counts
e_ce_t100	2000	0 / 4789	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
e_ce_t100	2001	0 / 4837	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
mask_ce_t100	2000	0 / 4789	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
mask_ce_t100	2001	0 / 4837	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
e_ce_t050	2000	0 / 4789	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
e_ce_t050	2001	0 / 4837	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
mask_ce_t050	2000	0 / 4789	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
mask_ce_t050	2001	0 / 4837	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}

Counts come from hash-validated final uint16 IDs and the saved editable bitmask; padding/framing are excluded. CSV text skips special tokens, so high strict chemical validity does not certify zero generated controls. These are final occurrences, not survival trajectories. The ideal clean-content forward endpoint at lambda 0.9 has approximately 9 ppm MASK probability at positive epsilon 1e-5; learned predictions, finite-step factorization, temperature and independent-prior initialization prevent using that as an empirical guarantee.

The manifest hashes every retained input, current imported helper source, archived generator and output. Large checkpoint bytes are not reloaded here: their digests and prior/CE/EMA acceptance are inherited from the bound CPU validation/training and independently rescored generation receipts. Original PDFs follow unchanged, beginning with the 43-page V9 overview.

GenMol / UDLM study through the paired objective evaluation

Completed V5/V6 screens, V7 resource pilot, V8 incident audit, separate V8b CE training and V9 molecular results

No superiority established. The highest observed repaired quality among the 22 disclosed engineering configurations is 57.031% (V6, s_t050_gibbs); contextual local MDLM quality is 85.8% and published GenMol V1 quality is 84.6%. This maximum is selected from small exploratory screens. A higher pilot mean, if present, does not establish superiority or authorize a final candidate promotion.

V9 clean CE minus CT repaired quality. primary at T=1.0: -5.500 percentage points; secondary at T=0.5: -2.000 percentage points. These are descriptive paired-seed differences; all strict/repaired metrics and every configuration remain disclosed below.

Study	Question and complete scope	Interpretation
V5	R/S/E x four temperatures; 24 runs / 1,536 requests	Historical 1,000-update batch-16 CT models; temperature exploration
V6	R/S/E x predictor/Gibbs; 12 runs / 768 requests	128 predictors versus 64 predictors + 64 Gibbs evaluations; mixed gains
V7	20-update batch-128 CT-E throughput pilot; 2,560 exposures	Resource feasibility only; no molecular evaluation or warm-start reuse
V8 / V8b	Separate 1,000-update CT and CE adaptations; 128,000 configured exposures each	Original CT controller/campaign failed; CT accepted separately; new CE controller completed
V9	CT/CE x temperatures 1.0 / 0.5; 8 runs / 800 requests	Primary CE minus CT at 1.0; informed secondary contrast at 0.5
Local MDLM	50,000 updates; 3 seeds / 3,000 requests	Contextual baseline; training/inference budgets and sample counts differ

New evidence and historical context. All 3,104 V5/V6/V9 molecular requests were independently rescored within their original runs; they are not pooled into a single model estimate. The historical 27-page report is appended without modifying its original bytes. Its CE-untrained sentence describes the earlier snapshot and is superseded by the completed V8b/V9 evidence here. Archived failure states remain failures.

Metrics. Validity = valid/requested; uniqueness = unique/valid; quality = unique valid molecules with QED >= 0.6 and SA <= 4, divided by requests; diversity is the released PyTDC fingerprint metric on unique valid molecules. Repaired decoding uses SAFE fix=True and largest-component selection by SMILES length; strict decoding uses fix=False and RDKit validation. Means and sample SD weight seeds equally; two-seed SD is not a confidence interval. Diversity and configurations are not pooled.

Training, controller outcomes and checkpoint semantics

Run	Updates / global batch / seed / GPUs	Training subprocess seconds	Accepted exposures / examples per second	Observed memory MiB
V7	20 / 128 / 1400 / 1	146.333	2,560 / 17.494	9637
V8_CT	1000 / 128 / 1500 / 2	785.822	not accepted by failed controller	9439, 11035
V8b_CE	1000 / 128 / 1500 / 2	706.595	128,000 / 181.150	11139, 9273

V8 CT incident. CT reached step 1,000 and exited zero, but the controller immediately found a residual distributed process group and failed, retaining both leases. CE never started in that campaign. A separately hashed CPU audit later verified the saved CT checkpoint and absent processes, then released the exact retained leases. It supports 128,000 configured exposures; the original accepted-exposure and throughput fields remain null. The exact transient group members were not captured, so the teardown-race explanation remains an inference.

Separate V8b CE run. Fresh MDLM EMA initialization preserved the original CE settings. A repaired controller allowed up to 15 seconds for teardown; its group exited after 0.751 seconds with 4 probes. Its own completion receipt accepted step 1,000, optimizer state, EMA updates, finite saved tensors and the CE marker/metadata. This does not convert the failed V8 campaign into a success.

Matched training. CT and CE each start fresh from MDLM 50k EMA, seed 1500, global batch 128 = 2 GPUs x microbatch 16 x accumulation 4; 1,000 updates = 128,000 configured exposures per arm. Both share the empirical prior with 0.0002 uniform mixture, FiLM, L1 schedule, full corruption alphabet and all-special-token clean-target mask. V7 and the earlier V5/V6 models used historical masks; V9 is not an isolated continuation of those screens.

Timing limits. Training-subprocess time includes startup and checkpoint saving. Short V7 startup overhead, one versus two GPUs, sequential runs and external processes prevent a controlled speedup claim. Memory values are two-second external aggregate GPU observations including other processes, not allocator peaks. Exposures are configured accounting, not distinct molecules or an independent live row trace. Saved-tensor finiteness does not prove every intermediate update was finite. CT/CE loss magnitudes are not comparable quality scores.

Checkpoint	SHA-256
V8 CT: separately audited	48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99
V8b CE: completed separately	b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1
Common warm-start MDLM 50k EMA	8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6

V9: all four objective settings

All V9 rows use 200 requests (two 100-request seeds, 1600/1601), EMA weights, 128 predictor evaluations, no Gibbs corrector, top-p 1.0, endpoint 1e-5, randomness 0 and min_add_len 40. CE clean logits are converted to LOO at the current noisy state/time before temperature and top-p; CT already predicts raw LOO. The conversion adds no backbone evaluation. Temperature 1.0 is primary; 0.5 was informed by V5/V6.

Config	Branch	Validity %	Uniqueness %	Quality %	Diversity
ct_t100	repaired	97.500 +/- 2.121	99.479 +/- 0.737	46.500 +/- 6.364	0.909 +/- 0.000
ct_t100	strict	39.000 +/- 5.657	100.000 +/- 0.000	21.000 +/- 1.414	0.886 +/- 0.007
ce_t100	repaired	98.000 +/- 1.414	99.485 +/- 0.729	41.000 +/- 4.243	0.903 +/- 0.004
ce_t100	strict	64.500 +/- 7.778	100.000 +/- 0.000	26.000 +/- 7.071	0.887 +/- 0.005
ct_t050	repaired	98.500 +/- 0.707	99.495 +/- 0.714	54.500 +/- 2.121	0.884 +/- 0.003
ct_t050	strict	77.000 +/- 1.414	100.000 +/- 0.000	44.000 +/- 1.414	0.868 +/- 0.008
ce_t050	repaired	99.000 +/- 0.000	99.495 +/- 0.714	52.500 +/- 4.950	0.888 +/- 0.006
ce_t050	strict	67.000 +/- 5.657	100.000 +/- 0.000	39.500 +/- 3.536	0.870 +/- 0.001

V9: paired CE minus CT effects

Positive deltas favor CE for that metric. Validity/uniqueness/quality deltas are percentage points; diversity uses its original scale. Values are the equal-seed mean +/- sample SD. An undefined metric withholds its summary until both declared seeds define it. The appendix retains each per-seed difference.

Contrast / T	Branch	Delta validity	Delta uniqueness	Delta quality	Delta diversity
primary / 1.0	repaired	0.500 +/- 0.707	0.005 +/- 0.008	-5.500 +/- 2.121	-0.006 +/- 0.004
primary / 1.0	strict	25.500 +/- 2.121	0.000 +/- 0.000	5.000 +/- 5.657	0.000 +/- 0.012
secondary / 0.5	repaired	0.500 +/- 0.707	0.000 +/- 1.428	-2.000 +/- 2.828	0.004 +/- 0.003
secondary / 0.5	strict	-10.000 +/- 7.071	0.000 +/- 0.000	-4.500 +/- 4.950	0.002 +/- 0.009

Seed pairing is not molecule-level matching: objectives and sampling paths differ. Both contrasts remain disclosed even when negative. The fixed design predates these checkpoint molecular outputs; earlier engineering results were already known, so this is not independent confirmation. Higher diversity alone does not imply higher validity or molecular quality. The appendix reports every requested metric under both decoding paths.

Reading the appendices. The first appendix is the complete paired V9 report with original per-run configuration, raw/rescore identities, CE helper provenance, generation counts and contrasts. The second is the preserved V5/V6 plus baseline overview, including all 18 historical settings and Gibbs ablations. The companion all_configurations.csv contains all 22 configuration means and SDs. The input manifest hashes the original reports, raw/run artifacts, training receipts/logs, CT audit and this generator. Large checkpoint identities are inherited from verified training/generation records; this overview does not reload the weights.

Equal updates/examples and shared hyperparameters do not imply equal compute or separately optimized objectives. All adaptations inherit the large MDLM pretraining budget; a matched extra-update MDLM control remains absent. Final UDLM seeds 0/1/2 remain reserved. No automatic promotion, further training or superiority declaration follows this report.

GenMol UDLM engineering exploration

engineering-v9-ct-ce-objectives | complete | 2026-09-07T11:25:39.800704+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

CT and clean CE share the training settings recorded in this protocol. Equal updates and examples do not imply equal compute or independently optimized objectives; training losses have different scales and targets.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED >= 0.6 and SA <= 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Predeclared contrasts subtract CT from CE within the same seed and decoding branch. Every declared contrast is retained, including negative outcomes. Paired mean and sample SD are withheld unless all declared seed pairs define that metric; unavailable pairs remain visible. Sample SD describes seed spread, not a confidence interval. Pairing seed labels does not imply coupled molecular trajectories.

Prospective sampling design

{"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "objective_comparison": {"contrast": "CE minus CT for each seed, then equal-seed mean; report both contrasts even when negative.", "primary": {"control_config": "ct_t100", "temperature": 1.0, "treatment_config": "ce_t100"}, "secondary": {"control_config": "ct_t050", "temperature": 0.5, "treatment_config": "ce_t050"}, "selection_history": "V5/V6 outcomes and V7 throughput were known before design; temperature 0.5 is an informed secondary engineering setting."}, "parameterization_controls": "Convert CE clean-denoiser logits to LOO at the current noisy state/time before temperature and top-p; CT logits already parameterize raw LOO.", "predictor_transitions_per_molecule": 128, "randomness": 0.0, "raw_loo_top_p": 1.0}

The original V8 CT controller and campaign remain failed. This comparison uses its separately audited checkpoint plus a new completed V8b CE run; never label V8 campaign success.

Training source revisions differ through controller/reporting changes; model/training implementation and resolved common training settings are matched by recorded hashes.

128000 training exposures are configured accounting, not distinct molecules or an independent live row trace. Finite final tensors do not establish finiteness of every intermediate update.

V5/V6 informed this engineering design, especially temperature 0.5; this is not independent confirmation or a superiority claim.

Two seeds and 100 requests per setting are small; pairing is by seed/run, not molecule-level trajectories or independence of generated observations.

Report validity, uniqueness, quality and diversity under both strict SAFE decoding and released repair/largest-component selection. Quality counts unique QED>=0.6 and SA<=4 molecules divided by requests.

Equal updates/examples and shared hyperparameters do not imply equal compute or independently optimized CT/CE objectives. The baseline uses 50k MDLM updates and three 1000-request seeds.

Final UDLM seeds 0/1/2 remain reserved. No automatic generation follows this materialization, and no model promotion or longer training follows the results.

Objective comparison

CT / raw-LOO predicts clean leave-one-out probabilities. Clean CE predicts clean denoiser probabilities, converted to LOO logits at the current noisy state and time before temperature or top-p. The conversion adds no backbone evaluation. Shared sampling settings do not imply independently optimized objectives. Objective labels and checkpoint evidence below describe completed runs only; the prospective design also covers pending runs.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"CE": {"checkpoint_sha256": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020}, "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b"}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size_bytes": 8621}, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}, "CT": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166}, "resolved_config_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}}, "common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846}, "original_v8_protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak_learning_rate": 0.0003, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50}, "training_implementation_hashes_sha256": "c9641b7e447d1f4617599e4df338fde0010fe68f7877566de37d2e6c5a6ccd2b", "training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdlm_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
ct_t100	1600	128	128	0
ct_t100	1601	128	128	0
ce_t100	1600	128	128	0
ce_t100	1601	128	128	0
ct_t050	1600	128	128	0
ct_t050	1601	128	128	0
ce_t050	1600	128	128	0
ce_t050	1601	128	128	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
ct_t100	1600	CT / raw-LOO	raw_loo	ema
ct_t100	1601	CT / raw-LOO	raw_loo	ema
ce_t100	1600	clean CE	x0_denoiser	ema
ce_t100	1601	clean CE	x0_denoiser	ema
ct_t050	1600	CT / raw-LOO	raw_loo	ema
ct_t050	1601	CT / raw-LOO	raw_loo	ema
ce_t050	1600	clean CE	x0_denoiser	ema

Config	Seed	Objective	Parameterization	Weights
ce_t050	1601	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_t100	2/2	0.9750 +/- 0.0212	0.9948 +/- 0.0074	0.4650 +/- 0.0636	0.9090 +/- 0.0001
ce_t100	2/2	0.9800 +/- 0.0141	0.9948 +/- 0.0073	0.4100 +/- 0.0424	0.9025 +/- 0.0036
ct_t050	2/2	0.9850 +/- 0.0071	0.9949 +/- 0.0071	0.5450 +/- 0.0212	0.8838 +/- 0.0028
ce_t050	2/2	0.9900 +/- 0.0000	0.9949 +/- 0.0071	0.5250 +/- 0.0495	0.8880 +/- 0.0059

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_t100	2/2	0.3900 +/- 0.0566	1.0000 +/- 0.0000	0.2100 +/- 0.0141	0.8864 +/- 0.0068
ce_t100	2/2	0.6450 +/- 0.0778	1.0000 +/- 0.0000	0.2600 +/- 0.0707	0.8867 +/- 0.0049
ct_t050	2/2	0.7700 +/- 0.0141	1.0000 +/- 0.0000	0.4400 +/- 0.0141	0.8683 +/- 0.0082
ce_t050	2/2	0.6700 +/- 0.0566	1.0000 +/- 0.0000	0.3950 +/- 0.0354	0.8700 +/- 0.0007

Predeclared paired CE minus CT contrasts

Differences use the same seed and decoding branch, with equal seed weights. Values are on the metric's 0-to-1 scale: 0.01 is one percentage point for quality. All declared pairs must define a metric before its mean and sample SD are shown. SD is seed spread, not a confidence interval. Seed labels do not imply paired molecules.

primary | temperature 1.0: ce_t100 minus ct_t100

released_comparable

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1600	+0.990000	+0.990000	+0.000000	completed

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1601	+0.960000	+0.970000	+0.010000	completed
uniqueness	1600	+1.000000	+1.000000	+0.000000	completed
uniqueness	1601	+0.989583	+0.989691	+0.000107	completed
quality	1600	+0.510000	+0.440000	-0.070000	completed
quality	1601	+0.420000	+0.380000	-0.040000	completed
diversity	1600	+0.909109	+0.900010	-0.009099	completed
diversity	1601	+0.908897	+0.905074	-0.003823	completed

Metric	Pairs defined	Mean CE minus CT	Sample SD
validity	2/2	+0.005000	0.007071
uniqueness	2/2	+0.000054	0.000076
quality	2/2	-0.055000	0.021213
diversity	2/2	-0.006461	0.003731

strict

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1600	+0.430000	+0.700000	+0.270000	completed
validity	1601	+0.350000	+0.590000	+0.240000	completed
uniqueness	1600	+1.000000	+1.000000	+0.000000	completed
uniqueness	1601	+1.000000	+1.000000	+0.000000	completed
quality	1600	+0.220000	+0.310000	+0.090000	completed
quality	1601	+0.200000	+0.210000	+0.010000	completed
diversity	1600	+0.891188	+0.883229	-0.007959	completed
diversity	1601	+0.881626	+0.890105	+0.008479	completed

Metric	Pairs defined	Mean CE minus CT	Sample SD
validity	2/2	+0.255000	0.021213
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.050000	0.056569
diversity	2/2	+0.000260	0.011624

secondary | temperature 0.5: ce_t050 minus ct_t050

released_comparable

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1600	+0.990000	+0.990000	+0.000000	completed
validity	1601	+0.980000	+0.990000	+0.010000	completed
uniqueness	1600	+0.989899	+1.000000	+0.010101	completed
uniqueness	1601	+1.000000	+0.989899	-0.010101	completed
quality	1600	+0.530000	+0.490000	-0.040000	completed
quality	1601	+0.560000	+0.560000	+0.000000	completed
diversity	1600	+0.885833	+0.892143	+0.006309	completed
diversity	1601	+0.881839	+0.883780	+0.001941	completed

Metric	Pairs defined	Mean CE minus CT	Sample SD
validity	2/2	+0.005000	0.007071
uniqueness	2/2	+0.000000	0.014285
quality	2/2	-0.020000	0.028284
diversity	2/2	+0.004125	0.003089

strict

Metric	Seed	CT	CE	CE minus CT	Pair status
validity	1600	+0.780000	+0.630000	-0.150000	completed
validity	1601	+0.760000	+0.710000	-0.050000	completed
uniqueness	1600	+1.000000	+1.000000	+0.000000	completed
uniqueness	1601	+1.000000	+1.000000	+0.000000	completed
quality	1600	+0.450000	+0.370000	-0.080000	completed
quality	1601	+0.430000	+0.420000	-0.010000	completed
diversity	1600	+0.874062	+0.869567	-0.004494	completed
diversity	1601	+0.862439	+0.870508	+0.008069	completed

Metric	Pairs defined	Mean CE minus CT	Sample SD
validity	2/2	-0.100000	0.070711
uniqueness	2/2	+0.000000	0.000000
quality	2/2	-0.045000	0.049497
diversity	2/2	+0.001787	0.008884

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
ct_t100	1600	completed	100	100	0.5100	0.2200	12.24
ct_t100	1601	completed	100	100	0.4200	0.2000	13.86
ce_t100	1600	completed	100	100	0.4400	0.3100	12.40
ce_t100	1601	completed	100	100	0.3800	0.2100	11.93
ct_t050	1600	completed	100	100	0.5300	0.4500	14.36
ct_t050	1601	completed	100	100	0.5600	0.4300	13.98
ce_t050	1600	completed	100	100	0.4900	0.3700	12.52
ce_t050	1601	completed	100	100	0.5600	0.4200	13.92

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: 718893198b0e08a2405e33177c64989499914207ff39b6880edbc5eff2b35c7d

v9-ct-t100 / seed 1600: {"checkpoint": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "7"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ct-t100/seed_1600/raw_samples.csv", "sha256": "f8b3994c1a65b0070cf248e44d795a4071ec06f521a7b693b8aba10d1457445d", "size_bytes": 30618}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ct-t100 / seed 1601: {"checkpoint": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay":

0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ct-t100/seed_1601/raw_samples.csv", "sha256": "13a78c24b80be0063ee9e44b9aace057d322c35acb2af114efef1f50cb374ee6", "size_bytes": 29880}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ce-t100 / seed 1600: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "7"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t100/seed_1600/raw_samples.csv", "sha256": "7680236fc4efc888b85c91569430d40ad0e44d9337983249957f9fd72a8b27c8", "size_bytes": 32879}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v9-ce-t100 / seed 1601: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256":

"f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1256}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 1279, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "6"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t100/seed_1601/raw_samples.csv", "sha256": "91de05669cdc5be79830e1c6173a395309ed478941d3e93cbd775556d52c84b3", "size_bytes": 32925}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v9-ct-t050 / seed 1600: {"checkpoint": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "7"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ct-t050/seed_1600/raw_samples.csv", "sha256": "11dbab23177d6591de03c5097edb9d7f4f7a000554f19906f07c043261cec716", "size_bytes": 32877}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ct-t050 / seed 1601: {"checkpoint": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1256}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 1279, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "6"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ct-t050/seed_1601/raw_samples.csv", "sha256": "13d0d3fb32634f20490389e74883f8ba88a196bb8cf565bdca0b5c8f0d9c028e", "size_bytes": 33609}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ce-t050 / seed 1600: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "7"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t050/seed_1600/raw_samples.csv", "sha256": "32b342171299c7bfd745c16007ade327ffdb67b94c14047749f140898356c205", "size_bytes": 32079}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v9-ce-t050 / seed 1601: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1256}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 1279, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "6"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t050/seed_1601/raw_samples.csv", "sha256": "41af77517633bd59388930f20f0435a3b57d86cb354a2e1b9ec5095a7f4e98dd", "size_bytes": 33179}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

GenMol / UDLM engineering study

Complete V5 temperature screen and V6 Gibbs comparison | source snapshot 2026-09-07T10:09:32.675735+00:00

No superiority established. No formal promotion or final candidate lock. The highest observed repaired quality across the 18 engineering configurations is 57.03% (V6, s_t050_gibbs), versus 85.80% for the frozen local MDLM baseline. This descriptive maximum is selected from a small screen; it is not a confirmatory estimate.

Evidence	Configurations / runs / requested samples	Seed and compute budget
V5 temperature screen	12 configurations; 24 runs; 1,536 samples, all independently rescored	R/S/E x temperatures 0.50, 0.70, 0.85, 1.00; seeds 1200/1201; 128 predictor NFEs
V6 Gibbs comparison	6 configurations; 12 runs; 768 samples, all independently rescored	R/S/E x predictor/Gibbs; seeds 1300/1301; temperature 0.50; 128 total NFEs
Frozen local MDLM baseline	1 configuration; 3 runs; 3,000 samples	Seeds 0/1/2; released confidence sampling; inference and training compute not matched

128 samples per engineering configuration versus 3,000 baseline samples. Every engineering result averages two 64-sample seeds. All 2,304 V5+V6 requests remain disclosed; they are not pooled to estimate a single model configuration. V6 temperature 0.50 was chosen using partial V5 results, so it is an informed engineering choice, not a separately selected global optimum.

Training and checkpoints. Each R/S/E model starts from the same 50,000-update MDLM EMA and adds 1,000 updates at global batch 16 (16,000 example exposures; one GPU; microbatch 2; accumulation 8; seed 17). All three retain their original checkpoint bytes. The local MDLM training used batch 2,046 for 50,000 updates (102.3 million requested exposures); the paper used batch 2,048. A matched extra-update MDLM control is still absent. Scheduler and FiLM choices were selected on E loss, then shared across arms.

Arm definitions. R uses released uniform UDLM with its known loss/forward schedule mismatch; S repairs that schedule using uniform categorical diffusion; E uses the same repaired process and empirical SAFE prior with a 0.0002 uniform mixture. None is a published reproduction of a fully optimized UDLM molecular model.

Metrics and uncertainty. Validity = valid/requested; uniqueness = unique/valid; quality = distinct valid molecules with QED >= 0.6 and SA <= 4, divided by requested samples. QED summarizes drug-likeness; SA is the synthetic-accessibility score (lower is easier). Diversity is the released PyTDC fingerprint diversity on unique valid molecules. Means and sample SD average seeds equally; diversity is not pooled. Two-seed SD is fragile, not a confidence interval or significance test; configuration selection and molecular dependence add uncertainty.

Decoding and next work. Repaired metrics use SAFE fix=True and select the largest component by SMILES string length. Strict metrics use fix=False plus RDKit validation, without repair or component selection. Final UDLM seeds 0/1/2 remain reserved (the historical baseline already used those seeds). CE denoising with checkpoint-bound CE-to-LOO conversion is implemented and CPU-tested, but remains untrained and has no molecular results in this report. The archived V4 failure remains failed and contributes no ranked configuration.

Every configuration: mean +/- sample SD across two seeds

V/U/Q are percentages; D is diversity. P = predictor backbone evaluation; G = fresh Gibbs evaluation. All engineering rows use 128 samples/configuration and top-p=1.00. Baseline rows are context, not matched tests.

Study / arm	T	Kernel	Repair V%	Repair U%	Repair Q%	Repair D	Strict V%	Strict U%	Strict Q%	Strict D
V5 / R	0.50	128P	99.22 +/- 1.10	100.00 +/- 0.00	48.44 +/- 4.42	0.881 +/- 0.001	63.28 +/- 5.52	100.00 +/- 0.00	29.69 +/- 4.42	0.869 +/- 0.001
V5 / R	0.70	128P	97.66 +/- 3.31	100.00 +/- 0.00	44.53 +/- 1.10	0.894 +/- 0.002	46.09 +/- 1.10	100.00 +/- 0.00	19.53 +/- 5.52	0.875 +/- 0.002
V5 / R	0.85	128P	96.88 +/- 2.21	99.21 +/- 1.12	35.94 +/- 6.63	0.911 +/- 0.002	39.84 +/- 3.31	100.00 +/- 0.00	21.09 +/- 5.52	0.877 +/- 0.010
V5 / R	1.00	128P	93.75 +/- 0.00	99.17 +/- 1.18	30.47 +/- 1.10	0.911 +/- 0.005	37.50 +/- 6.63	100.00 +/- 0.00	14.06 +/- 2.21	0.881 +/- 0.003
V5 / S	0.50	128P	97.66 +/- 1.10	99.19 +/- 1.14	41.41 +/- 14.36	0.897 +/- 0.004	60.16 +/- 1.10	100.00 +/- 0.00	23.44 +/- 8.84	0.878 +/- 0.004
V5 / S	0.70	128P	97.66 +/- 1.10	100.00 +/- 0.00	42.19 +/- 19.89	0.898 +/- 0.006	42.97 +/- 1.10	100.00 +/- 0.00	21.09 +/- 1.10	0.864 +/- 0.004
V5 / S	0.85	128P	96.09 +/- 1.10	100.00 +/- 0.00	50.78 +/- 5.52	0.904 +/- 0.004	39.84 +/- 5.52	100.00 +/- 0.00	24.22 +/- 3.31	0.871 +/- 0.004
V5 / S	1.00	128P	92.97 +/- 7.73	98.41 +/- 2.24	32.03 +/- 1.10	0.924 +/- 0.005	25.00 +/- 0.00	100.00 +/- 0.00	11.72 +/- 1.10	0.886 +/- 0.002
V5 / E	0.50	128P	97.66 +/- 1.10	100.00 +/- 0.00	45.31 +/- 6.63	0.892 +/- 0.006	47.66 +/- 1.10	100.00 +/- 0.00	22.66 +/- 5.52	0.878 +/- 0.002
V5 / E	0.70	128P	98.44 +/- 0.00	99.21 +/- 1.12	45.31 +/- 11.05	0.899 +/- 0.003	40.62 +/- 11.05	100.00 +/- 0.00	21.09 +/- 9.94	0.872 +/- 0.015
V5 / E	0.85	128P	94.53 +/- 1.10	96.68 +/- 2.38	39.84 +/- 3.31	0.904 +/- 0.005	32.03 +/- 3.31	100.00 +/- 0.00	13.28 +/- 3.31	0.883 +/- 0.010
V5 / E	1.00	128P	93.75 +/- 4.42	99.14 +/- 1.22	35.94 +/- 6.63	0.920 +/- 0.012	17.97 +/- 3.31	100.00 +/- 0.00	9.38 +/- 4.42	0.884 +/- 0.007
V6 / R	0.50	128P	100.00 +/- 0.00	98.44 +/- 2.21	53.91 +/- 5.52	0.885 +/- 0.007	65.62 +/- 8.84	100.00 +/- 0.00	36.72 +/- 3.31	0.874 +/- 0.001
V6 / R	0.50	64P+64G	97.66 +/- 1.10	100.00 +/- 0.00	47.66 +/- 3.31	0.893 +/- 0.012	61.72 +/- 7.73	100.00 +/- 0.00	32.03 +/- 5.52	0.877 +/- 0.008
V6 / S	0.50	128P	98.44 +/- 2.21	99.19 +/- 1.14	54.69 +/- 4.42	0.882 +/- 0.006	61.72 +/- 1.10	100.00 +/- 0.00	33.59 +/- 1.10	0.866 +/- 0.007
V6 / S	0.50	64P+64G	98.44 +/- 2.21	97.61 +/- 1.18	57.03 +/- 1.10	0.892 +/- 0.005	55.47 +/- 7.73	100.00 +/- 0.00	36.72 +/- 1.10	0.878 +/- 0.004
V6 / E	0.50	128P	96.88 +/- 0.00	99.19 +/- 1.14	46.88 +/- 8.84	0.896 +/- 0.010	53.91 +/- 5.52	100.00 +/- 0.00	31.25 +/- 2.21	0.887 +/- 0.006
V6 / E	0.50	64P+64G	98.44 +/- 2.21	98.41 +/- 0.04	51.56 +/- 6.63	0.889 +/- 0.003	55.47 +/- 9.94	100.00 +/- 0.00	29.69 +/- 0.00	0.874 +/- 0.003
Local MDLM	0.50	released	100.00 +/- 0.00	99.87 +/- 0.06	85.80 +/- 1.23	0.823 +/- 0.001	98.80 +/- 0.20	99.87 +/- 0.06	84.77 +/- 1.22	0.822 +/- 0.002

The local MDLM row averages three 1,000-sample seeds. Published GenMol V1 repaired reference: validity 100.0%, uniqueness 99.7%, quality 84.6%, diversity 0.818 (paper run means). Hardware, training, data provenance and sample counts differ; higher diversity alone does not establish better molecules.

Within-arm V6 Gibbs deltas at fixed total NFE

Treatment minus predictor control, pairing seed 1300 with 1300 and 1301 with 1301 at the same checkpoint. Control: 128 predictor evaluations. Treatment: 64 predictor transitions + 64 fresh random-scan Gibbs updates. The paired RNG streams diverge because Gibbs changes draw counts. These are descriptive paired-seed deltas, not per-molecule paired effects or a confidence interval. Temperature is 0.50 in every pair.

Arm	Decoding	Delta validity (pp)	Delta uniqueness (pp)	Delta quality (pp)	Delta diversity
R	repaired	-2.34 +/- 1.10	+1.56 +/- 2.21	-6.25 +/- 2.21	+0.008 +/- 0.019
R	strict	-3.91 +/- 16.57	+0.00 +/- 0.00	-4.69 +/- 2.21	+0.003 +/- 0.009
S	repaired	+0.00 +/- 4.42	-1.59 +/- 2.32	+2.34 +/- 3.31	+0.010 +/- 0.001
S	strict	-6.25 +/- 6.63	+0.00 +/- 0.00	+3.12 +/- 0.00	+0.012 +/- 0.011
E	repaired	+1.56 +/- 2.21	-0.78 +/- 1.18	+4.69 +/- 2.21	-0.007 +/- 0.007
E	strict	+1.56 +/- 15.47	+0.00 +/- 0.00	-1.56 +/- 2.21	-0.013 +/- 0.003

The means show mixed effects: E improves repaired quality but loses strict quality; S improves both quality means while strict validity falls; R declines in both quality and validity branches. There is no universal Gibbs gain. V5 and V6 use different seeds, so their differences are not paired comparisons.

A random-scan Gibbs step resamples one uniformly chosen editable coordinate per sequence and may leave its value unchanged. Exact stationarity requires true compatible LOO conditionals and untempered sampling; these learned predictions at temperature 0.50 form an approximate corrector. This comparison also trades away half the predictor transitions, so it evaluates that allocation, not adding free extra compute.

Checkpoint / budget	SHA-256 (original source bytes; not reserialized)
MDLM 50k EMA warm-start source	8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6
R: +1,000 updates / +16,000 exposures	d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c
S: +1,000 updates / +16,000 exposures	100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72
E: +1,000 updates / +16,000 exposures	dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005

Artifact trail. The input manifest hashes both report JSON/CSV/PDF files, all 36 run summaries and raw CSVs, both protocols and their configuration bytes, the three continuation training summaries, the local baseline aggregate and its verified PDF, and this generator. Checkpoint identities are inherited from the independently validated generation/training records; this overview does not reload multi-gigabyte checkpoints.

Unmodified appendices follow: complete independently rescored V5 report; complete independently rescored V6 report; frozen local MDLM baseline report. They retain original page labels and historical context. The companion CSV/JSON contain all configuration means, SDs, paired deltas, hashes and original identifiers. No significance test, superiority declaration, formal promotion, or reserved-seed UDLM generation occurred here.

GenMol UDLM engineering exploration

engineering-v5-temperature | complete | 2026-09-07T09:50:06.376168+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

The local MDLM model completed 50,000 updates; each UDLM arm starts from its EMA weights and adds 1,000 updates. Training compute is not matched.

The scheduler and FiLM conditioner were selected on E denoising loss and then shared by R/S/E. This is not independent optimization of each method.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED >= 0.6 and SA <= 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdlm_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_p100	2/2	0.9766 +/- 0.0110	1.0000 +/- 0.0000	0.4531 +/- 0.0663	0.8921 +/- 0.0055
e_t070_p100	2/2	0.9844 +/- 0.0000	0.9921 +/- 0.0112	0.4531 +/- 0.1105	0.8994 +/- 0.0028
e_t085_p100	2/2	0.9453 +/- 0.0110	0.9668 +/- 0.0238	0.3984 +/- 0.0331	0.9038 +/- 0.0054
e_t100_p100	2/2	0.9375 +/- 0.0442	0.9914 +/- 0.0122	0.3594 +/- 0.0663	0.9201 +/- 0.0115

Config	Seeds complete	validity	uniqueness	quality	diversity
s_t050_p100	2/2	0.9766 +/- 0.0110	0.9919 +/- 0.0114	0.4141 +/- 0.1436	0.8967 +/- 0.0043
s_t070_p100	2/2	0.9766 +/- 0.0110	1.0000 +/- 0.0000	0.4219 +/- 0.1989	0.8983 +/- 0.0064
s_t085_p100	2/2	0.9609 +/- 0.0110	1.0000 +/- 0.0000	0.5078 +/- 0.0552	0.9045 +/- 0.0044
s_t100_p100	2/2	0.9297 +/- 0.0773	0.9841 +/- 0.0224	0.3203 +/- 0.0110	0.9242 +/- 0.0053
r_t050_p100	2/2	0.9922 +/- 0.0110	1.0000 +/- 0.0000	0.4844 +/- 0.0442	0.8809 +/- 0.0010
r_t070_p100	2/2	0.9766 +/- 0.0331	1.0000 +/- 0.0000	0.4453 +/- 0.0110	0.8942 +/- 0.0019
r_t085_p100	2/2	0.9688 +/- 0.0221	0.9921 +/- 0.0112	0.3594 +/- 0.0663	0.9114 +/- 0.0020
r_t100_p100	2/2	0.9375 +/- 0.0000	0.9917 +/- 0.0118	0.3047 +/- 0.0110	0.9110 +/- 0.0051

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_p100	2/2	0.4766 +/- 0.0110	1.0000 +/- 0.0000	0.2266 +/- 0.0552	0.8785 +/- 0.0021
e_t070_p100	2/2	0.4062 +/- 0.1105	1.0000 +/- 0.0000	0.2109 +/- 0.0994	0.8718 +/- 0.0153
e_t085_p100	2/2	0.3203 +/- 0.0331	1.0000 +/- 0.0000	0.1328 +/- 0.0331	0.8833 +/- 0.0101
e_t100_p100	2/2	0.1797 +/- 0.0331	1.0000 +/- 0.0000	0.0938 +/- 0.0442	0.8839 +/- 0.0066
s_t050_p100	2/2	0.6016 +/- 0.0110	1.0000 +/- 0.0000	0.2344 +/- 0.0884	0.8781 +/- 0.0036
s_t070_p100	2/2	0.4297 +/- 0.0110	1.0000 +/- 0.0000	0.2109 +/- 0.0110	0.8641 +/- 0.0042
s_t085_p100	2/2	0.3984 +/- 0.0552	1.0000 +/- 0.0000	0.2422 +/- 0.0331	0.8713 +/- 0.0036
s_t100_p100	2/2	0.2500 +/- 0.0000	1.0000 +/- 0.0000	0.1172 +/- 0.0110	0.8859 +/- 0.0015
r_t050_p100	2/2	0.6328 +/- 0.0552	1.0000 +/- 0.0000	0.2969 +/- 0.0442	0.8686 +/- 0.0007
r_t070_p100	2/2	0.4609 +/- 0.0110	1.0000 +/- 0.0000	0.1953 +/- 0.0552	0.8749 +/- 0.0016
r_t085_p100	2/2	0.3984 +/- 0.0331	1.0000 +/- 0.0000	0.2109 +/- 0.0552	0.8772 +/- 0.0103
r_t100_p100	2/2	0.3750 +/- 0.0663	1.0000 +/- 0.0000	0.1406 +/- 0.0221	0.8811 +/- 0.0030

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_t050_p100	1200	completed	64	64	0.5000	0.2656	9.71
e_t050_p100	1201	completed	64	64	0.4062	0.1875	9.17
e_t070_p100	1200	completed	64	64	0.5312	0.2812	9.55
e_t070_p100	1201	completed	64	64	0.3750	0.1406	9.09
e_t085_p100	1200	completed	64	64	0.3750	0.1094	9.58
e_t085_p100	1201	completed	64	64	0.4219	0.1562	9.24
e_t100_p100	1200	completed	64	64	0.3125	0.0625	9.46
e_t100_p100	1201	completed	64	64	0.4062	0.1250	9.70
s_t050_p100	1200	completed	64	64	0.5156	0.2969	9.42
s_t050_p100	1201	completed	64	64	0.3125	0.1719	9.15
s_t070_p100	1200	completed	64	64	0.5625	0.2188	9.41
s_t070_p100	1201	completed	64	64	0.2812	0.2031	9.05
s_t085_p100	1200	completed	64	64	0.5469	0.2656	10.77
s_t085_p100	1201	completed	64	64	0.4688	0.2188	9.01
s_t100_p100	1200	completed	64	64	0.3281	0.1094	10.08
s_t100_p100	1201	completed	64	64	0.3125	0.1250	9.10
r_t050_p100	1200	completed	64	64	0.5156	0.3281	9.99
r_t050_p100	1201	completed	64	64	0.4531	0.2656	9.02
r_t070_p100	1200	completed	64	64	0.4531	0.2344	9.30
r_t070_p100	1201	completed	64	64	0.4375	0.1562	8.96
r_t085_p100	1200	completed	64	64	0.4062	0.2500	9.33
r_t085_p100	1201	completed	64	64	0.3125	0.1719	9.02
r_t100_p100	1200	completed	64	64	0.3125	0.1250	9.35
r_t100_p100	1201	completed	64	64	0.2969	0.1562	9.06

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: ecd4722c23deebe193aeb50c034122ca20e501cf7c379e4beb6514b4d496d1a7

v5-e-t050-p100 / seed 1200: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t050-p100/seed_1200/raw_samples.csv", "sha256": "575549aadb838cc83426a13fe3bc3233922f0f4b8ca79a618a6ba630e6e9f414", "size_bytes": 20244}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t050-p100 / seed 1201: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t050-p100/seed_1201/raw_samples.csv", "sha256": "8894501102bb1dc32ae64feec0d3e932dd7a24a364ae4ca6dc74d76db8e7febb", "size_bytes": 20232}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"} v5-e-t070-p100 / seed 1200: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t070-p100/seed_1200/raw_samples.csv", "sha256": "a0832e0b5489c32d25e9732b24e9160b32c48c485b39406f45d3a54c0ecc7cf8", "size_bytes": 20130}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t070-p100 / seed 1201: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t070-p100/seed_1201/raw_samples.csv", "sha256": "ab42728b2bea5adc25950a81f569a3df5c94b8f6a1ebdc8f585edd4631df541d", "size_bytes": 19412}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t085-p100 / seed 1200: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t085-p100/seed_1200/raw_samples.csv", "sha256": "06f0d2cfcf3215f70541d7c88c00cec89579b47d0975a8e99efdf35566240a8d", "size_bytes": 19118}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t085-p100 / seed 1201: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path":

"output/udlm/engineering_v5/v5-e-t085-p100/seed_1201/raw_samples.csv", "sha256": "327e7b74d31709973085a72743b62661abe2663f0969ea8ae44237ee6b1d88de", "size_bytes": 19054}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t100-p100 / seed 1200: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t100-p100/seed_1200/raw_samples.csv", "sha256": "c5441a763ac9843d4a0865bbb0f33db9e47a9a1c55d2195de1bc29d1bbde9e51", "size_bytes": 18276}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-e-t100-p100 / seed 1201: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-e-t100-p100/seed_1201/raw_samples.csv", "sha256": "dd15322742447031c8d04ef61c37f6e0a98f46f32c57ff4e50a258b32ef20693", "size_bytes": 18294}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t050-p100 / seed 1200: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t050-p100/seed_1200/raw_samples.csv", "sha256": "67750d50a016eae3431aa95d3ec47a4976e1c6295323bc69777bcefa49d4caf9", "size_bytes": 21059}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t050-p100 / seed 1201: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy":

{"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t050-p100/seed_1201/raw_samples.csv", "sha256": "1de893cf2f402a2021f8eb08dd6e9eca5a55e3627ada5dc54b990c1e7fe04af2", "size_bytes": 21005}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t070-p100 / seed 1200: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t070-p100/seed_1200/raw_samples.csv", "sha256": "ca74886bf6be3cee9a7a51f43724141fb63bd9ffd26b6ebc5a99c658d6cb8290", "size_bytes": 19956}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t070-p100 / seed 1201: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t070-p100/seed_1201/raw_samples.csv", "sha256": "fc3769c7bcf5071f1b8984559bb86f1abac89b8c07b47828c0770b8d2081b298", "size_bytes": 19596}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t085-p100 / seed 1200: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 5, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t085-p100/seed_1200/raw_samples.csv", "sha256": "db612b3231bc4f17e07b052863303215c6de0d5930e8a8114adced6f1ed89f7d", "size_bytes": 19991}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t085-p100 / seed 1201: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t085-p100/seed_1201/raw_samples.csv", "sha256": "86c26ea024fb46143df0f5de9c67801411e5489c2a6db8dad7e37064c4e2b947", "size_bytes": 19489}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t100-p100 / seed 1200: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t100-p100/seed_1200/raw_samples.csv", "sha256": "6356b28ba7eeea9c2128ac0add101b0705c8f408b2309963798fa71fb282be8e", "size_bytes": 18533}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-s-t100-p100 / seed 1201: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-s-t100-p100/seed_1201/raw_samples.csv", "sha256": "f56974a534d8a39435858a77c3c341c459780fb6f4dda83798bd4ab55430ae34", "size_bytes": 18519}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t050-p100 / seed 1200: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t050-p100/seed_1200/raw_samples.csv", "sha256": "f941d5b4cc21edeef012814945ec2ac534d0d5ab655336e93ef4ac1b821168e6", "size_bytes": 21718}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t050-p100 / seed 1201: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t050-p100/seed_1201/raw_samples.csv", "sha256": "d27ee7308b3596eb46dae8c165aafa22ca96aeb21135a6411d7ab030c7ce2b28", "size_bytes": 20790}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t070-p100 / seed 1200: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t070-p100/seed_1200/raw_samples.csv", "sha256": "bbeb58d2bbb6faeece43f13fc7a21d2204137f61406eac9a48e7f352d42cd820", "size_bytes": 20237}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t070-p100 / seed 1201: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.7}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t070-p100/seed_1201/raw_samples.csv", "sha256": "f9b92cd25a1cff261b49f516fc350bda523480b5320e59c8f2b259bcf3b0adc6", "size_bytes": 20340}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t085-p100 / seed 1200: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t085-p100/seed_1200/raw_samples.csv", "sha256": "81d10fb2e12e415f8c54fb1471ddbd2a22242a90a9b18c2b8e672a0edc05ce72", "size_bytes": 19906}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t085-p100 / seed 1201: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.85}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t085-p100/seed_1201/raw_samples.csv", "sha256": "76cb87c29b6008868ad7a92ee99bc1d266030d4290f0b344185c8f99aa5f89b0", "size_bytes": 19645}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t100-p100 / seed 1200: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t100-p100/seed_1200/raw_samples.csv", "sha256": "a6a569384cc07e077edc60ce7e3883a0b4866a62d64024a6526dad5dea81dfa3", "size_bytes": 19414}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

v5-r-t100-p100 / seed 1201: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v5/v5-r-t100-p100/seed_1201/raw_samples.csv", "sha256": "a1bb88ced33cd8c2b6716b415fcfe2a83e54cd7d9eb2012d676be09ff54f3049", "size_bytes": 19966}, "source": "12e13ddf9b1323bbfb60fe289b5dd363ca0c3c05"}

GenMol UDLM engineering exploration

engineering-v6-gibbs | complete | 2026-09-07T10:09:32.675735+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

The local MDLM model completed 50,000 updates; each UDLM arm starts from its EMA weights and adds 1,000 updates. Training compute is not matched.

The scheduler and FiLM conditioner were selected on E denoising loss and then shared by R/S/E. This is not independent optimization of each method.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED >= 0.6 and SA <= 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Prospective sampling design

{"arms": ["R", "S", "E"], "claim_boundary": "Learned/tempered LOO conditionals are approximate; exact Gibbs stationarity is established only for the toy true-conditional tests.", "corrector_coordinate_selection": "one uniformly selected editable coordinate per row, fresh prediction at the post-predictor state and time", "frozen_before_v6_generation": true, "gibbs_treatment": {"corrector_updates": 64, "nfe": 128, "predictor_transitions": 64}, "partial_v5_observed_before_design": "E and S temperature outcomes through 0.85; no final evaluation seeds. Temperature 0.5 is fixed across arms to favor syntactic fidelity, not selected as the global quality winner.", "predictor_control": {"corrector_updates": 0, "nfe": 128, "predictor_transitions": 128}, "temperature": 0.5, "top_p": 1.0}

Exploratory parameter selection, not a confirmatory benchmark.

Only 1000 UDLM continuation updates: matched method training superiority is not isolated.

Historical baseline has 1000 samples per seed versus 64 here; diversity and uniqueness vary with sample count.

Strict molecular validity is reported separately from repaired validity.

The corrector changes transition allocation at fixed total NFE, not training.

Temperature 0.5 is a common engineering choice informed by partial v5 syntax diagnostics.

Corrector predictions can depend on their own noisy token; exact stationarity is not asserted for the learned model.

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdlm_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
e_t050_predictor	1300	128	128	0
e_t050_predictor	1301	128	128	0
e_t050_gibbs	1300	128	64	64
e_t050_gibbs	1301	128	64	64
s_t050_predictor	1300	128	128	0
s_t050_predictor	1301	128	128	0
s_t050_gibbs	1300	128	64	64
s_t050_gibbs	1301	128	64	64
r_t050_predictor	1300	128	128	0
r_t050_predictor	1301	128	128	0
r_t050_gibbs	1300	128	64	64
r_t050_gibbs	1301	128	64	64

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_predictor	2/2	0.9688 +/- 0.0000	0.9919 +/- 0.0114	0.4688 +/- 0.0884	0.8959 +/- 0.0102
e_t050_gibbs	2/2	0.9844 +/- 0.0221	0.9841 +/- 0.0004	0.5156 +/- 0.0663	0.8886 +/- 0.0033

Config	Seeds complete	validity	uniqueness	quality	diversity
s_t050_predictor	2/2	0.9844 +/- 0.0221	0.9919 +/- 0.0114	0.5469 +/- 0.0442	0.8822 +/- 0.0064
s_t050_gibbs	2/2	0.9844 +/- 0.0221	0.9761 +/- 0.0118	0.5703 +/- 0.0110	0.8921 +/- 0.0052
r_t050_predictor	2/2	1.0000 +/- 0.0000	0.9844 +/- 0.0221	0.5391 +/- 0.0552	0.8850 +/- 0.0070
r_t050_gibbs	2/2	0.9766 +/- 0.0110	1.0000 +/- 0.0000	0.4766 +/- 0.0331	0.8932 +/- 0.0124

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_predictor	2/2	0.5391 +/- 0.0552	1.0000 +/- 0.0000	0.3125 +/- 0.0221	0.8874 +/- 0.0056
e_t050_gibbs	2/2	0.5547 +/- 0.0994	1.0000 +/- 0.0000	0.2969 +/- 0.0000	0.8744 +/- 0.0029
s_t050_predictor	2/2	0.6172 +/- 0.0110	1.0000 +/- 0.0000	0.3359 +/- 0.0110	0.8659 +/- 0.0067
s_t050_gibbs	2/2	0.5547 +/- 0.0773	1.0000 +/- 0.0000	0.3672 +/- 0.0110	0.8776 +/- 0.0045
r_t050_predictor	2/2	0.6562 +/- 0.0884	1.0000 +/- 0.0000	0.3672 +/- 0.0331	0.8743 +/- 0.0009
r_t050_gibbs	2/2	0.6172 +/- 0.0773	1.0000 +/- 0.0000	0.3203 +/- 0.0552	0.8769 +/- 0.0080

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_t050_predictor	1300	completed	64	64	0.5312	0.3281	9.11
e_t050_predictor	1301	completed	64	64	0.4062	0.2969	7.44
e_t050_gibbs	1300	completed	64	64	0.5625	0.2969	9.46
e_t050_gibbs	1301	completed	64	64	0.4688	0.2969	8.86
s_t050_predictor	1300	completed	64	64	0.5156	0.3281	9.25
s_t050_predictor	1301	completed	64	64	0.5781	0.3438	8.94
s_t050_gibbs	1300	completed	64	64	0.5625	0.3594	9.59
s_t050_gibbs	1301	completed	64	64	0.5781	0.3750	8.90

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
r_t050_predictor	1300	completed	64	64	0.5781	0.3906	9.24
r_t050_predictor	1301	completed	64	64	0.5000	0.3438	8.90
r_t050_gibbs	1300	completed	64	64	0.5000	0.3594	9.31
r_t050_gibbs	1301	completed	64	64	0.4531	0.2812	8.81

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: b50d97dcdd10f35c85aba61c9b9b948081ae9726809abb25b6c9e0a8834a92e9

v6-e-t050-predictor / seed 1300: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index":

"5"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-predictor/seed_1300/raw_samples.csv", "sha256":
"e65e1f806e958601528d3c48b31e727f004318f427956ba7dd77ceece5d3b5d2", "size_bytes": 20407}, "sampling_budget": {"corrector_steps_per_molecule": 0,
"gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-predictor / seed 1301: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm",
"exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256":
"f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0,
"softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay":
0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one
full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256":
"f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0,
"randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true,
"single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe":
{"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python",
"used_memory_mib": 872}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 895, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid":
"GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus",
"max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison":
"strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index":
"6"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-predictor/seed_1301/raw_samples.csv", "sha256":
"30f4f1668bd4716728fe9efed2adbe5efc397a709c75716a17f57c8f0c57b167", "size_bytes": 20096}, "sampling_budget": {"corrector_steps_per_molecule": 0,
"gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-gibbs / seed 1300: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm",
"exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256":
"f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0,
"softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol":
{"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema":
{"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128,
"nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total
predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256":
"f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0,
"randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true,
"single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe":
{"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python",
"used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid":
"GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus",
"max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison":
"strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index":
"2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-gibbs/seed_1300/raw_samples.csv", "sha256":
"7912f9c622d82870edbb211c1688a73fb100b0cefd49e8a614a8622ae88db94b", "size_bytes": 19526}, "sampling_budget": {"corrector_steps_per_molecule": 64,
"gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-gibbs / seed 1301: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm",
"exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256":
"f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0,

"softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-gibbs/seed_1301/raw_samples.csv", "sha256": "64c88ce12d8ac7a022d6b1715cbef0ffa2ec0434145bae6f9995895d94eb492f", "size_bytes": 20839}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-predictor / seed 1300: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-predictor/seed_1300/raw_samples.csv", "sha256": "cfe608dd00991b6b95db83694542150cc55feeabfa262953d9a312f317ea3017", "size_bytes": 20527}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-predictor / seed 1301: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes":

[{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-predictor/seed_1301/raw_samples.csv", "sha256": "df3e72494039fd6a19e51489cfdea60f69d74bea55deee8f94c44b6d19c20ff0", "size_bytes": 20731}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-gibbs / seed 1300: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-gibbs/seed_1300/raw_samples.csv", "sha256": "3d389848d51f373a2f911fb06c34048f23276cef913b5acb201495d40cfdb0d1", "size_bytes": 20711}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-gibbs / seed 1301: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name":

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v6-r-t050-gibbs / seed 1300: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-gibbs/seed_1300/raw_samples.csv", "sha256": "82a3ed576ca875885098408c9279a0b75d1d33d4590b7d82bcb0ea23d8f32823", "size_bytes": 20943}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-r-t050-gibbs / seed 1301: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-gibbs/seed_1301/raw_samples.csv", "sha256": "a7f427c0741144239f64a594a7cca57bdbb98d8297cf62a2dbfb7065105dc678", "size_bytes": 20300}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

AUDITABLE 3 x 1,000 DE NOVO EVALUATION

GenMol from-scratch benchmark

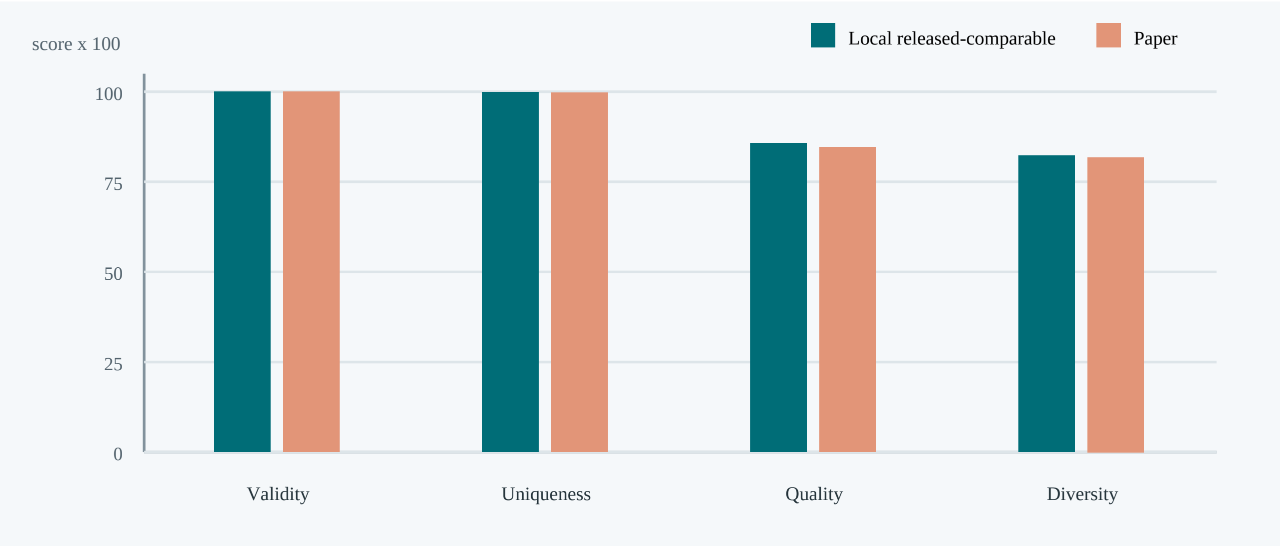
Final 50,000-step checkpoint compared with published GenMol V1 Table 1

COMPARATIVE RESULT - NOT AN EXACT REPRODUCTION CLAIM. All three local seeds completed with 1,000 retained raw rows and passed independent count, checkpoint, configuration, and artifact-hash validation.

Headline comparison

Metric	Local released-comparable	Published GenMol V1	Local - paper
Validity	100.00 +/- 0.00%	100.00 +/- 0.00%	+0.00 pp
Uniqueness	99.87 +/- 0.06%	99.70 +/- 0.10%	+0.17 pp
Quality	85.80 +/- 1.23%	84.60 +/- 0.80%	+1.20 pp
Diversity	0.8230 +/- 0.0013	0.8180 +/- 0.0010	+0.0050
Generation time	91.44 +/- 11.58 s	21.10 +/- 0.40 s	+70.34 s

Headline values are the arithmetic mean and sample SD of the three seed-level metrics. SD uses sqrt(sum((x_i - mean)^2)/(n-1)), n=3 (ddof=1); rows are not pooled before computing headline rates.



Bars show means on a 0-100 scale (diversity multiplied by 100). Uncertainty remains in the tables to avoid suggesting paired estimates.

Interpretation boundary

Differences are descriptive. The local checkpoint was trained from scratch with documented protocol deviations, and the published run seeds are undisclosed. No significance, parity, superiority, or exact reproduction conclusion follows from this table.

Seed-level results and decoding funnel

The released-comparable branch is the Table 1 comparison target. The strict branch exposes output that decodes without SAFE repair or largest-component selection.

Released-comparable metrics

Seed	Validity	Uniqueness	Quality	Diversity	Generation
0	100.00% (1000/1000)	99.80% (998/1000)	84.90% (849/1000)	0.8239 (n=998)	103.84 s
1	100.00% (1000/1000)	99.90% (999/1000)	87.20% (872/1000)	0.8215 (n=999)	80.92 s
2	100.00% (1000/1000)	99.90% (999/1000)	85.30% (853/1000)	0.8237 (n=999)	89.55 s
Mean +/- sample SD	100.00 +/- 0.00%	99.87 +/- 0.06%	85.80 +/- 1.23%	0.8230 +/- 0.0013	91.44 +/- 11.58 s

Strict metrics

Seed	Validity	Uniqueness	Quality	Diversity
0	99.00% (990/1000)	99.80% (988/990)	83.70% (837/1000)	0.8236 (n=988)
1	98.60% (986/1000)	99.90% (985/986)	86.10% (861/1000)	0.8206 (n=985)
2	98.80% (988/1000)	99.90% (987/988)	84.50% (845/1000)	0.8232 (n=987)
Mean +/- sample SD	98.80 +/- 0.20%	99.87 +/- 0.06%	84.77 +/- 1.22%	0.8225 +/- 0.0016

Strict vs repaired funnel (counts)

Seed	Requested / raw SAFE	Strict valid > unique > quality	Repair recovered / released lost	Released valid > unique > quality	Largest component applied
0	1000 / 1000	990 > 988 > 837	10 / 0	1000 > 998 > 849	14
1	1000 / 1000	986 > 985 > 861	14 / 0	1000 > 999 > 872	14
2	1000 / 1000	988 > 987 > 845	12 / 0	1000 > 999 > 853	13
Sum	3000 / 3000	2964 > 2960 > 2543	36 / 0	3000 > 2996 > 2574	41

Unique counts are summed within seed; molecules are not deduplicated across independent runs.

Exact metric definitions

Metric/path	Definition and denominator
Released Comparable	Released-code-compatible path: repair SAFE with fix=True, canonicalize, retain the largest disconnected component, then evaluate.
Strict	Strict path: canonical SAFE decode with fix=False and no largest-component selection. It is a diagnostic local addition and has no Table 1 counterpart.
Validity	valid_count / 1,000 requested samples, independently per seed.
Uniqueness	first-occurrence unique valid molecules / valid_count, per seed.
Quality	first-occurrence unique valid molecules with QED >= 0.6 and SA <= 4.0 / 1,000 requested samples, per seed.
Diversity	TDC Diversity on first-occurrence unique valid molecules; diversity_input_count records the input size.
Generation Time	Released-compatible wall-clock boundary: one 1,000-sample model/tokenizer call plus SAFE fix=True repair and largest-component selection. It equals model_sampling_and_tokenizer + released_postprocessing; model loading, strict decoding, and metric evaluation are excluded.

Checkpoint and protocol provenance

Checkpoint identity: global step 50,000; SHA-256 8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6. All three summaries agree.

Verified checkpoint

Field	Value
Resolved path	/home/aidar.alimbayev/Documents/genmolv2/outputs/paper_v1/checkpoints/50000.ckpt
SHA-256	8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6
Size	1,396,998,679 bytes
Global step	50,000
EMA	202 finite shadow tensors; 50,000 updates; decay 0.9999
Config SHA-256	42eadcb450cfc97f151157cae54d2a12ee88227d7368a9834b2fb75e40d40830
Effective-config SHA-256	8b6bcbb62baff2b1837627ff1109fd3d20a73fc900e2d9b2dcb1f1b36b67213d
Sampling-config SHA-256	ee1c7815ed56d51b311ea4bab2c88100b5bdf2897c6ddb987d406e5b3096aec4
Runner SHA-256	a77d7c84d8628403c6d7a11edebce5bf9b8405e3f8b79a326a6e06bd7f444cea
Sampler source SHA-256	30b2e04282440471fb9059c35fa51ad4adcda9f7a81c838e2b5ca62073d1111f
Model source SHA-256	7e6004fbb0cc22f68988d76d52b0c6066194a8a2e7a182eaadbf73dd856402ab
Chemistry utilities SHA-256	0c7de4c7225f80e9a8cbbdb99018e5e8555c114ce6deea7dfcd593fc5e0f3867
Data utilities SHA-256	7d707bdef8edd6c5235935ca867ba7d799a00e0624a5ea9662665bfa2a94daa9
Bracket converter SHA-256	d3dca11fc12d7f4f0f439c03fc7d497f4efc6d87091388b7031c9d856279a618
Length distribution	n=249,455; min/median/max=10/49/87; SHA-256 9795cde72c60e9e58cd6dc99551801bdf47d00de6e060cbafb5f30584b19cf27

Audited evaluation protocol

Setting	Audited local value
Runs / seeds	3 independent invocations; explicit seeds 0, 1, 2
Samples	1,000 per seed; one generation batch; 3,000 requested total
Softmax temperature	0.5
Randomness	0.5
Minimum added length	40
Confidence reveal count	N=1, implicit MDLM.get_num_steps_confidence default used by released sampler
Released postprocessing	SAFE fix=True; canonical decode; retain component with maximum SMILES string length
Strict diagnostic	SAFE fix=False; no component selection
Generation timer	model_sampling_and_tokenizer + released_postprocessing

Loaded tokenizer provenance

Tokenizer field	Recorded inference value
Identity	requested=datamol-io/safe-gpt; runtime name=not retained; class transformers.tokenization_utils_fast.PreTrainedTokenizerFast
Revision	declared=not declared; resolved=not recorded
Vocabulary	base=1,880; effective=1,882 after repository-added tokens; SHA-256 8fdf74b22e1fa097c83b45022ddd1270adb45873962b0b5b3aba480e0ff8260f
Added/backend hashes	added=818312ee2fd79eb194e7c1e9677d671c46e8f6e13e0371cc6c01a4df196c241a; backend=not serializable
Special IDs	PAD=3, BOS=1, EOS=2, MASK=4

Documented deviations and published source

Known training and evaluation deviations

Item	Local	Published/reference	Why it matters
Training global batch size	2,046 (62 microbatch x 3 GPUs x 11 accumulation)	2,048	Small protocol deviation; checkpoint is not byte-identical to the released model.
Training hardware	3 NVIDIA RTX A6000 GPUs; 46:19:39 wall time	8 NVIDIA A100 GPUs; approximately 5 hours	Training and inference wall times are hardware-dependent and not speed reproductions.
Dataset and tokenizer revisions	datamol-io/safe-gpt identifiers were unpinned	Exact local revisions cannot be proven from this run	Possible corpus/tokenizer drift remains an uncontrolled reproducibility variable.
Evaluation seeds	Explicit seeds 0, 1, 2	Three runs; seed values undisclosed	Runs are repeatable locally but cannot be paired seed-for-seed with Table 1.
Training RNG seed	Not recorded in the Hydra training config or retained training log	Not disclosed	The training trajectory cannot be recreated seed-for-seed.

HARDWARE TIMING CAVEAT. Local generation time is measured on user-selected, shared RTX A6000 inference devices, using the released de_novo_generation timer boundary (model/tokenizer + SAFE repair + largest component). Each launch snapshot satisfied an exclusive utilization threshold strictly below 15% and the recorded free-memory minimum, but active compute processes were allowed. Table 1 used an A100 and a different software environment. Sharing can affect runtime, so the time delta is descriptive and is not a speed claim.

Published source

Primary paper PDF: https://raw.githubusercontent.com/mlresearch/v267/main/assets/lee25o/lee25o.pdf. Published values transcribed here: validity 100.0 +/- 0.0%, uniqueness 99.7 +/- 0.1%, quality 84.6 +/- 0.8%, diversity 0.818 +/- 0.001, and time 21.1 +/- 0.4 seconds.

The paper reports three 1,000-sample runs, but its seed values are undisclosed. Local seeds 0, 1, and 2 are therefore explicit repeatability choices, not matched replicas of paper runs.

Audit trail and claims boundary

Per-seed artifact identities

Seed	Raw CSV SHA-256	Summary SHA-256	Inference device
0	fb1a255f04092fdf8763b18eccb866eef02f1d7dde422727b00ce549bc7b2b2f	c8b312dd905ddd0f972920c2851569a4c2468c98aa46dfa2ac3f4dcbb3567a65	NVIDIA RTX A6000; physical 2, logical 0
1	5f20bd6731e6c095189de1f7c82ecd46123115b308b1b232c167d64afd1c6dbb	f95f8261b37f82a711848c18b7b15ebea1235858099b2bf8ef5d53b528699575	NVIDIA RTX A6000; physical 2, logical 0
2	70c962d46685855b8a1ac099e834956c32a1f0d8dda4a60146b9a8dbb1cac58a	a5c1be00950744b634c23ced6d2fea2f49ffe75b0d1419b3b01b9841bd3c4983	NVIDIA RTX A6000; physical 2, logical 0

Shared low-utilization GPU launch snapshots

Seed	User-selected physical ID	Point-in-time load / free memory	Active compute processes at selection
0	2 (allowed: 2)	5% (< 15%); 47,387 MiB free (minimum 30,000)	1: PID 1701406 /home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python (1730 MiB)
1	2 (allowed: 2)	5% (< 15%); 47,387 MiB free (minimum 30,000)	1: PID 1701406 /home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python (1730 MiB)
2	2 (allowed: 2)	5% (< 15%); 47,387 MiB free (minimum 30,000)	1: PID 1701406 /home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python (1730 MiB)

Validation performed before aggregation

Exactly seeds 0/1/2; completed summary markers; top-level/nested seed and sample-count agreement; 1,000 ordered CSV rows per seed; exact CSV schema; raw-file SHA-256; expected checkpoint hash/size/global step; exact V1 sampling/generation protocol; distinct ordered raw outputs; complete RNG seeding; user-selected physical-ID provenance and UUID to logical cuda:0 mapping; shared-GPU snapshots satisfying the recorded exclusive utilization and minimum-free-memory policy, with process inventories preserved; cross-seed dependency, tokenizer, implementation, checkpoint, effective-config, and runner identity; recomputed strict/released validity, first-occurrence uniqueness, quality gates, repair recovery, largest-component flags, failure counts, ratios, and timing invariants.

Caveats

1. This is a from-scratch implementation benchmark against published reference values, not a claim of exact model or environment reproduction.

2. The released-comparable path intentionally repairs malformed SAFE and may drop disconnected components. Strict results expose the effect; neither path should be silently substituted for the other.

3. With three local seeds, SD is descriptive. No hypothesis test, confidence interval, or equivalence claim is made. Paper seed values are unavailable.

4. Descriptive only: the local timer matches the released de_novo_generation boundary (model/tokenizer plus SAFE repair and largest-component selection), but local inference uses user-selected shared RTX A6000 hardware while the paper used an A100. Point-in-time selection below the recorded exclusive utilization threshold does not prevent concurrent workload interference, and the software environments are not identical.

5. Unpinned in the local training configuration; exact dataset and tokenizer revisions are unproven.

6. The local training RNG seed was not recorded. Evaluation seeds are explicit, but they do not repair that training-provenance gap.

7. Implementation hashes establish that all local seeds used the same code. This reporter does not assert that those hashes equal a pristine NVIDIA release checkout.

BOTTOM LINE. This document faithfully describes the completed local benchmark and its distance from the published headline values. It does not assert that the released checkpoint, original seeds, training data revision, hardware, or full environment were reproduced exactly.

Generated at 2026-09-05T19:53:15.353694+00:00. Machine-readable JSON and CSV are emitted in the caller-selected aggregate directory.

GenMol UDLM engineering exploration

engineering-v10-predictor-resolution | complete | 2026-09-07T11:40:48.728679+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

CT and clean CE share the training settings recorded in this protocol. Equal updates and examples do not imply equal compute or independently optimized objectives; training losses have different scales and targets.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED >= 0.6 and SA <= 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Prospective sampling design

{"compute_comparison": "512 predictor NFEs cost four times 128 per molecule; measure wall time and make no equal-compute or fixed-NFE claim.", "corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "nfe_by_configuration": {"ce_t050_n128": 128, "ce_t050_n512": 512, "ct_t050_n128": 128, "ct_t050_n512": 512}, "parameterization_controls": "Convert CE clean-denoiser logits to LOO at the current noisy state/time before temperature and top-p; CT logits already parameterize raw LOO.", "randomness": 0.0, "raw_loo_top_p": 1.0, "resolution_comparison": {"aggregation": "Equal-seed mean and sample SD; all declared seed pairs must define the metric before summary; undefined values are not zero.", "contrasts": [{"contrast_id": "ct_512_minus_128", "control_config": "ct_t050_n128", "method": "CT", "treatment_config": "ct_t050_n512"}, {"contrast_id": "ce_512_minus_128", "control_config": "ce_t050_n128", "method": "CE", "treatment_config": "ce_t050_n512"}], "decoding_branches": ["released_comparable", "strict"], "direction": "512 minus 128 within each method and seed; report both contrasts even when negative.", "metrics": ["validity", "uniqueness", "quality", "diversity"], "paired_unit": "Seed/run labels, not molecule-level trajectories."}, "selection_history": "V5/V6 and complete V9 outcomes were observed before this design. Temperature 0.5 is an informed engineering choice; both frozen CT and CE checkpoints remain included.", "temperature": 0.5}

This design follows V5/V6 and complete V9 results; temperature 0.5 was selected using those outcomes and is not independent confirmation.

The 512-step treatment costs four times the predictor NFEs of 128, so any gain is not an equal-compute win; wall-clock scaling is measured.

The finite-state oracle uses exact untempered conditionals; it does not prove an improvement for learned temperature-0.5 molecular models or identify the cause of their quality gap.

The original V8 CT controller and campaign remain failed. This panel reuses its separately audited checkpoint plus the completed V8b CE checkpoint; neither is retrained or substituted.

Training source revisions differ through controller/reporting changes; model/training implementation and resolved common training settings match by recorded hashes.

128000 training exposures are configured accounting, not distinct molecules or an independent live row trace. Finite final tensors do not establish finiteness of every intermediate update.

Two seeds of 100 requests per setting are small; pairing is by seed/run, not molecule-level trajectories or independence. Withhold each contrast summary until both pairs define its metric.

Report validity, uniqueness, quality and diversity under strict SAFE decoding and released repair/largest-component selection, plus every per-seed 512-minus128 contrast. Quality counts unique QED>=0.6 and SA<=4 molecules divided by requests.

MDLM and paper means remain contextual comparisons with different sample/training budgets. Final UDLM seeds 0/1/2 remain reserved; no superiority, promotion or additional experiment is authorized by the results alone.

Objective comparison

CT / raw-LOO predicts clean leave-one-out probabilities. Clean CE predicts clean denoiser probabilities, converted to LOO logits at the current noisy state and time before temperature or top-p. The conversion adds no backbone evaluation. Shared sampling settings do not imply independently optimized objectives. Objective labels and checkpoint evidence below describe completed runs only; the prospective design also covers pending runs.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"CE": {"checkpoint_sha256": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020}, "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b"}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size_bytes": 8621}, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}, "CT": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166}, "resolved_config_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}}, "common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846}, "original_v8_protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak_learning_rate": 0.0003, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant":

"empirical_frequency", "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50}, "training_implementation_hashes_sha256": "c9641b7e447d1f4617599e4df338fde0010fe68f7877566de37d2e6c5a6ccd2b", "training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdlm_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
ct_t050_n128	1700	128	128	0
ct_t050_n128	1701	128	128	0
ct_t050_n512	1700	512	512	0
ct_t050_n512	1701	512	512	0
ce_t050_n128	1700	128	128	0
ce_t050_n128	1701	128	128	0
ce_t050_n512	1700	512	512	0
ce_t050_n512	1701	512	512	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
ct_t050_n128	1700	CT / raw-LOO	raw_loo	ema
ct_t050_n128	1701	CT / raw-LOO	raw_loo	ema
ct_t050_n512	1700	CT / raw-LOO	raw_loo	ema
ct_t050_n512	1701	CT / raw-LOO	raw_loo	ema

Config	Seed	Objective	Parameterization	Weights
ce_t050_n128	1700	clean CE	x0_denoiser	ema
ce_t050_n128	1701	clean CE	x0_denoiser	ema
ce_t050_n512	1700	clean CE	x0_denoiser	ema
ce_t050_n512	1701	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_t050_n128	2/2	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.4550 +/- 0.0778	0.8918 +/- 0.0064
ct_t050_n512	2/2	0.9950 +/- 0.0071	1.0000 +/- 0.0000	0.4500 +/- 0.0283	0.8907 +/- 0.0011
ce_t050_n128	2/2	0.9900 +/- 0.0141	0.9949 +/- 0.0072	0.4650 +/- 0.0071	0.8928 +/- 0.0026
ce_t050_n512	2/2	0.9850 +/- 0.0071	1.0000 +/- 0.0000	0.4800 +/- 0.0424	0.8927 +/- 0.0011

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_t050_n128	2/2	0.7200 +/- 0.0990	1.0000 +/- 0.0000	0.3550 +/- 0.1202	0.8754 +/- 0.0016
ct_t050_n512	2/2	0.7350 +/- 0.0212	1.0000 +/- 0.0000	0.3250 +/- 0.0354	0.8764 +/- 0.0052
ce_t050_n128	2/2	0.7250 +/- 0.0636	1.0000 +/- 0.0000	0.3600 +/- 0.0000	0.8781 +/- 0.0007
ce_t050_n512	2/2	0.7150 +/- 0.0071	1.0000 +/- 0.0000	0.3850 +/- 0.0354	0.8779 +/- 0.0051

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
ct_t050_n128	1700	completed	100	100	0.4000	0.2700	15.14
ct_t050_n128	1701	completed	100	100	0.5100	0.4400	13.75

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
ct_t050_n512	1700	completed	100	100	0.4300	0.3000	57.41
ct_t050_n512	1701	completed	100	100	0.4700	0.3500	52.18
ce_t050_n128	1700	completed	100	100	0.4700	0.3600	14.95
ce_t050_n128	1701	completed	100	100	0.4600	0.3600	14.05
ce_t050_n512	1700	completed	100	100	0.5100	0.4100	63.67
ce_t050_n512	1701	completed	100	100	0.4500	0.3600	53.26

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: cc6e3ced657f6ec66f611398e65b8b0e79598b3e76acbef77c72cfc4a7d0c4fd

v10-ct-t050-n128 / seed 1700: {"checkpoint": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid":

"GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ct-t050-n128/seed_1700/raw_samples.csv", "sha256": "c6abc1958624063f0bd300358d32d7a4d2ede63d90a989c95a17591577b18f5a", "size_bytes": 33423}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e"}

v10-ct-t050-n128 / seed 1701: {"checkpoint": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1286}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 1309, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ct-t050-n128/seed_1701/raw_samples.csv", "sha256": "f4c29b56c1603787ff0d24435090db94b3b43f45b69d2da8bd97da381b19b20a", "size_bytes": 33293}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e"}

v10-ct-t050-n512 / seed 1700: {"checkpoint": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ct-t050-n512/seed_1700/raw_samples.csv", "sha256": "2cecd4957084f9510054f48a72ce65ec46520625a205b28c690424f3c7215e4a", "size_bytes": 34079}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e"}

v10-ct-t050-n512 / seed 1701: {"checkpoint": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1286}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 1309, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ct-t050-n512/seed_1701/raw_samples.csv", "sha256": "e0482deaa9678cb658d05679c4d0688c5ceb7aa3dd1edcbc4460cb9e49ace906", "size_bytes": 32825}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e"}

v10-ce-t050-n128 / seed 1700: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1286}, {"pid": 508363, "process_name": "/home/aidar.alimbayev/Documents/genmolv2/.venv/bin/python", "used_memory_mib": 2434}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 3749, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": 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"c17e848eb46896aff191ecd535d6a3cfa757c37e", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}} v10-ce-t050-n128 / seed 1701: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ce-t050-n128/seed_1701/raw_samples.csv", "sha256": "aaafbe612da92ecefb46a04460ab18a4f53828e5ffd8f2f25a48037ce9aa0973", "size_bytes": 32214}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v10-ce-t050-n512 / seed 1700: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1286}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 1309, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ce-t050-n512/seed_1700/raw_samples.csv", "sha256": "84ea6a2ffcc9644055669221cfa1c69a58cd5a81d68d1e74d5e3d7fca67d6c88", "size_bytes": 33955}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v10-ce-t050-n512 / seed 1701: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "7"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ce-t050-n512/seed_1701/raw_samples.csv", "sha256": "2035f6cb04278cf8d3c7950d3396be103a4d38cadeac21baa2ff44335a0cf7a8", "size_bytes": 32638}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

V10: sampling resolution within CT and CE

512 predictor evaluations minus 128, separately within each method; temperature 0.5, seeds 1700/1701, 100 requests per seed. Four configurations / eight runs / 800 requests, all independently rescored. This is a resolution contrast, not CE minus CT and not equal compute: 512 uses four times the model evaluations.

No superiority established. V5/V6 and complete V9 outcomes informed this design. The separately audited V8 CT and completed V8b CE checkpoints are reused without retraining. The original V8 controller/campaign remains failed. Final UDLM seeds 0/1/2 remain reserved. Local MDLM quality 85.8% and paper GenMol V1 84.6% are contextual, differently budgeted comparisons.

Configuration	Branch	Validity %	Uniqueness %	Quality %	Diversity
ct_t050_n128	repaired	100.000 +/- 0.000	100.000 +/- 0.000	45.500 +/- 7.778	0.89176 +/- 0.00644
ct_t050_n128	strict	72.000 +/- 9.899	100.000 +/- 0.000	35.500 +/- 12.021	0.87538 +/- 0.00160
ct_t050_n512	repaired	99.500 +/- 0.707	100.000 +/- 0.000	45.000 +/- 2.828	0.89072 +/- 0.00114
ct_t050_n512	strict	73.500 +/- 2.121	100.000 +/- 0.000	32.500 +/- 3.536	0.87636 +/- 0.00515
ce_t050_n128	repaired	99.000 +/- 1.414	99.490 +/- 0.722	46.500 +/- 0.707	0.89277 +/- 0.00262
ce_t050_n128	strict	72.500 +/- 6.364	100.000 +/- 0.000	36.000 +/- 0.000	0.87814 +/- 0.00067
ce_t050_n512	repaired	98.500 +/- 0.707	100.000 +/- 0.000	48.000 +/- 4.243	0.89265 +/- 0.00113
ce_t050_n512	strict	71.500 +/- 0.707	100.000 +/- 0.000	38.500 +/- 3.536	0.87789 +/- 0.00508

Per-configuration means +/- sample SD weight seeds equally; no pooling across configurations. Validity and quality divide by requests; uniqueness divides by valid molecules. Quality counts unique valid molecules with QED >= 0.6 and SA <= 4. Diversity is the released PyTDC metric on unique valid molecules. Repaired decoding uses SAFE fix=True and largest-component selection by SMILES length; strict uses fix=False plus RDKit validation.

Paired molecular changes: 512 minus 128

Method / seed	Branch	Delta validity pp	Delta uniqueness pp	Delta quality pp	Delta diversity
CT / 1700	repaired	0.000	0.000	3.000	-0.00640
CT / 1701	repaired	-1.000	0.000	-4.000	0.00433
CT mean +/- SD	repaired	-0.500 +/- 0.707	0.000 +/- 0.000	-0.500 +/- 4.950	-0.00104 +/- 0.00759
CT / 1700	strict	10.000	0.000	3.000	-0.00153
CT / 1701	strict	-7.000	0.000	-9.000	0.00349
CT mean +/- SD	strict	1.500 +/- 12.021	0.000 +/- 0.000	-3.000 +/- 8.485	0.00098 +/- 0.00355
CE / 1700	repaired	-1.000	0.000	4.000	0.00093
CE / 1701	repaired	0.000	1.020	-1.000	-0.00117
CE mean +/- SD	repaired	-0.500 +/- 0.707	0.510 +/- 0.722	1.500 +/- 3.536	-0.00012 +/- 0.00149
CE / 1700	strict	-5.000	0.000	5.000	-0.00337
CE / 1701	strict	3.000	0.000	0.000	0.00287
CE mean +/- SD	strict	-1.000 +/- 5.657	0.000 +/- 0.000	2.500 +/- 3.536	-0.00025 +/- 0.00441

Pairing is by seed/run labels, not molecule-level trajectories: different step counts change random draws and reverse paths. All declared pairs must define a metric before its mean/SD is shown; undefined is never zero. Both methods and negative changes remain disclosed. Two-seed SD is descriptive, not a confidence interval or significance test.

Generation wall time: 512 / 128

Method / seed	128-step seconds	512-step seconds	Ratio / mean +/- sample SD
CT / 1700	15.143	57.415	3.792
CT / 1701	13.748	52.184	3.796
CT mean +/- SD	--	--	3.794 +/- 0.003
CE / 1700	14.948	63.671	4.259
CE / 1701	14.052	53.255	3.790
CE mean +/- SD	--	--	4.025 +/- 0.332

Generation seconds include model sampling/tokenizer decoding plus released SAFE postprocessing; they exclude checkpoint loading and metric scoring. The mean is the equal-seed mean of paired ratios, not a ratio of means. Sequential GPU scheduling, device mapping and external activity can change wall time; four times NFE need not mean four times wall time.

Raw SAFE ring and parenthesis diagnostics

Configuration / seed	Requests	Odd ring label	Bad parentheses	Both flags	Either flag	Flagged strict-valid
ct_t050_n128 / 1700	100	19	14	3	30	0
ct_t050_n128 / 1701	100	8	12	2	18	0
ct_t050_n512 / 1700	100	7	14	1	20	0
ct_t050_n512 / 1701	100	11	11	1	21	0
ce_t050_n128 / 1700	100	12	8	3	17	0
ce_t050_n128 / 1701	100	12	14	4	22	0
ce_t050_n512 / 1700	100	12	12	2	22	0
ce_t050_n512 / 1701	100	11	11	2	20	0

Odd ring label: remove bracket atoms, recognize single-digit, %NN and %(integer) labels, normalize integer labels and flag odd counts. Even counts allow legitimate label reuse and do not establish chemical validity. Parentheses: remove bracket atoms and %(integer) labels, then flag a negative prefix depth or nonzero final depth. Counts overlap.

Method: mean 512-minus-128 count +/- SD	Odd ring	Parentheses	Both	Either	Flagged strict-valid
CT	-4.500 +/- 10.607	-0.500 +/- 0.707	-1.500 +/- 0.707	-3.500 +/- 9.192	0.000 +/- 0.000
CE	-0.500 +/- 0.707	0.500 +/- 4.950	-1.500 +/- 0.707	1.500 +/- 4.950	0.000 +/- 0.000

Released repair behavior: counts across the two 100-request seeds

Configuration	Requests	Recovered: count / requests (%)	Largest-component selection: count / requests (%)
ct_t050_n128	200	56 / 200 (28.0%)	45 / 200 (22.5%)
ct_t050_n512	200	52 / 200 (26.0%)	40 / 200 (20.0%)
ce_t050_n128	200	53 / 200 (26.5%)	39 / 200 (19.5%)
ce_t050_n512	200	54 / 200 (27.0%)	42 / 200 (21.0%)

Recovered means released decoding produced a valid selected molecule where strict decoding did not. Largest-component selection is the saved released-selection flag. Both fractions divide by all requests, not only valid molecules, and can overlap. CSV/JSON preserve the per-seed counts; these descriptive counts do not isolate a causal effect of repair.

Lexical flags are limited diagnostics, not a complete SAFE/SMILES parser or a causal explanation of quality. Each count uses all 100 requested rows, including decode failures and repeats; no deduplication across seeds or configurations. Fewer flags need not improve QED, SA, validity or quality. The finite-state oracle motivated resolution testing but does not prove gains for learned, temperature-0.5 molecular models.

The original complete V10 report remains unchanged and contains all configurations, generation identities, device mappings and per-run metrics. This appendix hashes that report, its PDF/CSV, raw rows, summaries, controllers, prospective/prior evidence and the exact reused lexical/helper source. JSON/CSV retain exact values and counts. No new molecular sampling, chemistry rescoring, checkpoint load or automatic promotion occurred here.

GenMol UDLM engineering exploration

engineering-v12-ce-mask-prior | complete | 2026-09-07T12:56:25.414190+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

Both prior arms use clean-token CE and the training settings recorded in this protocol. Changing the corruption prior changes reconstruction difficulty; equal updates and examples do not imply equal compute, and lower training loss does not establish better molecular generation.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED >= 0.6 and SA <= 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Predeclared contrasts subtract empirical CE from MASK-rich CE within the same seed and decoding branch. Every declared contrast is retained, including negative outcomes. Paired mean and sample SD are withheld unless all declared seed pairs define that metric; unavailable pairs remain visible. Sample SD describes seed spread, not a confidence interval. Pairing seed labels does not imply coupled molecular trajectories.

Prospective sampling design

{"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "parameterization_controls": "Both CE arms convert clean-denoiser logits to LOO at current state/time before temperature and top-p; no extra model call.", "predictor_transitions_per_molecule": 128, "prior_comparison": {"primary": {"control_config": "e_ce_t100", "temperature": 1.0, "treatment_config": "mask_ce_t100"}, "secondary": {"control_config": "e_ce_t050", "temperature": 0.5, "treatment_config": "mask_ce_t050"}}, "randomness": 0.0, "raw_loo_top_p": 1.0}

V9/V10 and the frozen MDLM transfer diagnostic informed this exploratory prior comparison; no superiority or independent confirmation claim.

Both arms use clean CE; changed corruption and reverse laws change reconstruction difficulty. Equal updates/exposures do not imply equal runtime.

Original V11 failed before training and remains failed. V11b is a separate successful fresh attempt; original V8 CT remains failed with a separate checkpoint audit and is context only.

Two seeds of 100 requests are small. Pair by seed, not molecular trajectory. Report every configuration and failure, both decoding branches, all four metrics and signed MASK-minus-empirical contrasts.

Quality counts unique valid QED>=0.6 and SA<=4 molecules divided by all requests. Report recovery and largest-component selection frequencies with denominators.

Disclose final editable MASK counts and their editable-position denominator from summary sampled_token_control_audit; decoded strings skip specials and cannot supply this count.

128000 configured training exposures are not necessarily distinct molecules. Finite final checkpoints do not establish finiteness of every update.

Local MDLM .858 and paper GenMol V1 .846 use different training/sample budgets and are context only; no automatic promotion or extra training. Final seeds0/1/2 remain reserved.

Prior comparison

Both arms predict clean-denoiser probabilities using CE training, with empirical versus MASK-rich empirical stationary priors. Both convert to LOO logits at the current noisy state and time before temperature or top-p, with no extra backbone evaluation. Changing the prior changes the training corruption and reverse law. Checkpoint evidence below describes completed runs only; the prospective design also covers pending runs.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"E_CE": {"checkpoint_sha256": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "checkpoint_size_bytes": 1636492999, "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020}, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b", "size_bytes": 3161}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size_bytes": 8621}, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}, "udlm_prior_metadata": {"active_token_ids_sha256": "2fd62050919fdf6ee578df881733e47834f662406c219a1f247116e04396a5ae", "active_vocab_size": 1880, "antithetic_sampling": true, "comparison_role": "empirical_prior_treatment", "excluded_token_ids": [], "frequency_active_token_count": 517090, "frequency_artifact_path": "experiments/udlm/token_frequency/train_first_10000.json", "frequency_artifact_schema_version": 1, "frequency_artifact_sha256": "088c78e75611f3cc42c4011e1da6f65a377e673b9cba07a28b126b0fc62f06ed", "frequency_content_token_count": 517090, "frequency_dataset_repo_id": "datamol-io/safe-gpt", "frequency_dataset_revision": "b83175cd7394e7a4027478a35b2f9d1dda3ac62f", "frequency_dataset_selection": "first 10000 streaming rows", "frequency_dataset_split": "train", "frequency_example_count": 10000, "frequency_implementation_git_sha": "56a96b2cd02f9be648a641c51c3d2d8b1ff3033b", "frequency_ordered_text_sha256": "53aee8e5592fc96159788e86519abbbcc9f1ab7c6348a1cb59a939bd57051d8f", "full_vocab_size": 1880, "noise_eps": 0.001, "objective_scope": "model_dependent_ct_integrand_without_parameter_independent_endpoint_kl", "prior_source": "pinned_frequency_artifact_uniform_mixture", "process_family": "rank_one_continuous_categorical", "sampling_eps": 0.001, "schedule_variant": "schedule_consistent_residual_forward_and_loss", "schema_version": 1, "stationary_probs_sha256": "3b6ce79449e0abc083bc860b3543be554667435ce9f7f1b92c0f960ff27400c3", "tokenizer_json_sha256": "0db5f4dbdc7e8ff759e98483759611a426e187ee7f3f0a91edc8800abe7bf140", "tokenizer_repo_id": "datamol-io/safe-gpt", "tokenizer_revision": "3d5fa0988383e898d5ac5db7cd52bf715bc37061", "uniform_mixture_weight": 0.0002, "variant": "empirical_frequency"}}, "MASK_CE": {"checkpoint_sha256": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "checkpoint_size_bytes": 1636493702, "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/launch_manifest.json", "sha256": "457912bcf1975f0adb72cb2b5f02ec07c32915dc2119f184c19d6003edb60f85", "size_bytes": 24826}, "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v11b_mask_prior.json", "sha256": "5ecf638fa8fc7a3707a0497c8a358697610f52c8af7f2d6468458890e26368eb", "size_bytes": 7982}, "request_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/request_manifest.json", "sha256": "a039045f0e3589b3c15ac56fa215cb27bf0bde36b87902c0e1028c456b45f956", "size_bytes": 14003}, "resolved_config_sha256":

"5d73af507e81b643cf8f35225ab5d796357024d1665d039fdf40b5f37df15698", "source_revision": "be18a3244a717978e027d6e43ba0ac20b8137e4a", "terminal_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/terminal_manifest.json", "sha256": "891f5d6d609146a02ebb18caa7176bf555e3f67fe2493795c8a3e1fccb264779", "size_bytes": 17667}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}, "udlm_prior_metadata": {"active_token_ids_sha256": "2fd62050919fdf6ee578df881733e47834f662406c219a1f247116e04396a5ae", "active_vocab_size": 1880, "antithetic_sampling": true, "base_stationary_probs_sha256": "3b6ce79449e0abc083bc860b3543be554667435ce9f7f1b92c0f960ff27400c3", "comparison_role": "mask_rich_empirical_prior_treatment", "excluded_token_ids": [], "frequency_active_token_count": 517090, "frequency_artifact_path": "experiments/udlm/token_frequency/train_first_10000.json", "frequency_artifact_schema_version": 1, "frequency_artifact_sha256": "088c78e75611f3cc42c4011e1da6f65a377e673b9cba07a28b126b0fc62f06ed", "frequency_content_token_count": 517090, "frequency_dataset_repo_id": "datamol-io/safe-gpt", "frequency_dataset_revision": "b83175cd7394e7a4027478a35b2f9d1dda3ac62f", "frequency_dataset_selection": "first 10000 streaming rows", "frequency_dataset_split": "train", "frequency_example_count": 10000, "frequency_implementation_git_sha": "56a96b2cd02f9be648a641c51c3d2d8b1ff3033b", "frequency_ordered_text_sha256": "53aee8e5592fc96159788e86519abbbcc9f1ab7c6348a1cb59a939bd57051d8f", "full_vocab_size": 1880, "mask_mixture_weight": 0.9, "mask_token_id": 4, "noise_eps": 0.001, "objective_scope": "model_dependent_ct_integrand_without_parameter_independent_endpoint_kl", "prior_source": "mask_point_mass_mixture_with_pinned_smoothed_frequency", "process_family": "rank_one_continuous_categorical", "sampling_eps": 0.001, "schedule_variant": "schedule_consistent_residual_forward_and_loss", "schema_version": 1, "stationary_probs_sha256": "07d011c07bdac8fdbe096f514ded6e2d1b6850268c48bec8dd0e3f3853d013eb", "tokenizer_json_sha256": "0db5f4dbdc7e8ff759e98483759611a426e187ee7f3f0a91edc8800abe7bf140", "tokenizer_repo_id": "datamol-io/safe-gpt", "tokenizer_revision": "3d5fa0988383e898d5ac5db7cd52bf715bc37061", "uniform_mixture_weight": 0.0002, "variant": "mask_rich_empirical"}}}, "common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_definition": "Empirical-arm configuration after removing parameterization and normalizing callback.dirpath. The MASK arm matches after restoring empirical_frequency and removing mask_mixture_weight.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846}, "original_v11_failure": {"relative_path": "output/udlm/engineering_v11/mask_ce_1000_b128_w2/terminal_manifest.json", "sha256": "6a4aa47e1f7b76ad5efc3ce03cd3e4a55c0db4d95778b0cf60cc55cab7207408", "size_bytes": 14510}, "original_v8_ct_context": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166}, "resolved_config_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}, "original_v8_protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak_learning_rate": 0.0003, "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50}, "training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdlm_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
e_ce_t100	2000	128	128	0
e_ce_t100	2001	128	128	0
mask_ce_t100	2000	128	128	0
mask_ce_t100	2001	128	128	0
e_ce_t050	2000	128	128	0
e_ce_t050	2001	128	128	0
mask_ce_t050	2000	128	128	0
mask_ce_t050	2001	128	128	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
e_ce_t100	2000	clean CE	x0_denoiser	ema
e_ce_t100	2001	clean CE	x0_denoiser	ema
mask_ce_t100	2000	clean CE	x0_denoiser	ema
mask_ce_t100	2001	clean CE	x0_denoiser	ema
e_ce_t050	2000	clean CE	x0_denoiser	ema
e_ce_t050	2001	clean CE	x0_denoiser	ema
mask_ce_t050	2000	clean CE	x0_denoiser	ema
mask_ce_t050	2001	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_ce_t100	2/2	0.9900 +/- 0.0141	0.9950 +/- 0.0071	0.4250 +/- 0.0636	0.9008 +/- 0.0034
mask_ce_t100	2/2	0.9950 +/- 0.0071	0.9950 +/- 0.0071	0.5000 +/- 0.0566	0.9046 +/- 0.0004
e_ce_t050	2/2	0.9800 +/- 0.0141	1.0000 +/- 0.0000	0.4850 +/- 0.0212	0.8945 +/- 0.0074
mask_ce_t050	2/2	0.9950 +/- 0.0071	0.9949 +/- 0.0071	0.5400 +/- 0.0283	0.8863 +/- 0.0042

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_ce_t100	2/2	0.5850 +/- 0.0636	1.0000 +/- 0.0000	0.2750 +/- 0.0212	0.8854 +/- 0.0061
mask_ce_t100	2/2	0.5850 +/- 0.0354	1.0000 +/- 0.0000	0.3400 +/- 0.0566	0.8823 +/- 0.0007
e_ce_t050	2/2	0.7150 +/- 0.0071	1.0000 +/- 0.0000	0.3750 +/- 0.0354	0.8830 +/- 0.0079
mask_ce_t050	2/2	0.7650 +/- 0.0495	1.0000 +/- 0.0000	0.4200 +/- 0.0424	0.8765 +/- 0.0003

Predeclared paired MASK minus empirical contrasts

Differences use the same seed and decoding branch, with equal seed weights. Values are on the metric's 0-to-1 scale: 0.01 is one percentage point for quality. All declared pairs must define a metric before its mean and sample SD are shown. SD is seed spread, not a confidence interval. Seed labels do not imply paired molecules.

primary | temperature 1.0: mask_ce_t100 minus e_ce_t100

released_comparable

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
validity	2000	+1.000000	+1.000000	+0.000000	completed
validity	2001	+0.980000	+0.990000	+0.010000	completed
uniqueness	2000	+0.990000	+0.990000	+0.000000	completed
uniqueness	2001	+1.000000	+1.000000	+0.000000	completed
quality	2000	+0.380000	+0.540000	+0.160000	completed

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
quality	2001	+0.470000	+0.460000	-0.010000	completed
diversity	2000	+0.898434	+0.904817	+0.006383	completed
diversity	2001	+0.903207	+0.904317	+0.001110	completed

Metric	Pairs defined	Mean MASK minus empirical	Sample SD
validity	2/2	+0.005000	0.007071
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.075000	0.120208
diversity	2/2	+0.003746	0.003729

strict

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
validity	2000	+0.630000	+0.610000	-0.020000	completed
validity	2001	+0.540000	+0.560000	+0.020000	completed
uniqueness	2000	+1.000000	+1.000000	+0.000000	completed
uniqueness	2001	+1.000000	+1.000000	+0.000000	completed
quality	2000	+0.260000	+0.380000	+0.120000	completed
quality	2001	+0.290000	+0.300000	+0.010000	completed
diversity	2000	+0.881115	+0.881820	+0.000705	completed
diversity	2001	+0.889762	+0.882860	-0.006902	completed

Metric	Pairs defined	Mean MASK minus empirical	Sample SD
validity	2/2	+0.000000	0.028284
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.065000	0.077782
diversity	2/2	-0.003099	0.005379

secondary | temperature 0.5: mask_ce_t050 minus e_ce_t050

released_comparable

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
validity	2000	+0.990000	+1.000000	+0.010000	completed
validity	2001	+0.970000	+0.990000	+0.020000	completed
uniqueness	2000	+1.000000	+1.000000	+0.000000	completed
uniqueness	2001	+1.000000	+0.989899	-0.010101	completed
quality	2000	+0.500000	+0.560000	+0.060000	completed
quality	2001	+0.470000	+0.520000	+0.050000	completed
diversity	2000	+0.889283	+0.883360	-0.005924	completed
diversity	2001	+0.899687	+0.889340	-0.010346	completed

Metric	Pairs defined	Mean MASK minus empirical	Sample SD
validity	2/2	+0.015000	0.007071
uniqueness	2/2	-0.005051	0.007142
quality	2/2	+0.055000	0.007071
diversity	2/2	-0.008135	0.003127

strict

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
validity	2000	+0.720000	+0.800000	+0.080000	completed
validity	2001	+0.710000	+0.730000	+0.020000	completed
uniqueness	2000	+1.000000	+1.000000	+0.000000	completed
uniqueness	2001	+1.000000	+1.000000	+0.000000	completed
quality	2000	+0.400000	+0.450000	+0.050000	completed

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
quality	2001	+0.350000	+0.390000	+0.040000	completed
diversity	2000	+0.877424	+0.876253	-0.001171	completed
diversity	2001	+0.888618	+0.876684	-0.011935	completed

Metric	Pairs defined	Mean MASK minus empirical	Sample SD
validity	2/2	+0.050000	0.042426
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.045000	0.007071
diversity	2/2	-0.006553	0.007611

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_ce_t100	2000	completed	100	100	0.3800	0.2600	15.24
e_ce_t100	2001	completed	100	100	0.4700	0.2900	15.45
mask_ce_t100	2000	completed	100	100	0.5400	0.3800	14.85
mask_ce_t100	2001	completed	100	100	0.4600	0.3000	15.63
e_ce_t050	2000	completed	100	100	0.5000	0.4000	14.87
e_ce_t050	2001	completed	100	100	0.4700	0.3500	15.11
mask_ce_t050	2000	completed	100	100	0.5600	0.4500	15.24
mask_ce_t050	2001	completed	100	100	0.5200	0.3900	15.38

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32,

'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: f4ea756fabb9e5133c72f302e8398bcbb8d0eabaa81f183b017af8c19b02cadd

v12-e-ce-t100 / seed 2000: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-e-ce-t100/seed_2000/raw_samples.csv", "sha256": "2f7aeed34d62a4ce8285287468400755d6300e73a6262c00e4e117a72e5701e9", "size_bytes": 32935}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-e-ce-t100 / seed 2001: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-e-ce-t100/seed_2001/raw_samples.csv", "sha256": "473c9897f2f72f9b7c93d2d5225c032e82c6c5e9855851fca6654b1fbc8c1b18", "size_bytes": 31685}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-mask-ce-t100 / seed 2000: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-mask-ce-t100/seed_2000/raw_samples.csv", "sha256": "7f618bc2e4f2424bc675a5dfa331a83b934935310e88ba7771899e16e3fe60ce", "size_bytes": 31923}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-mask-ce-t100 / seed 2001: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-mask-ce-t100/seed_2001/raw_samples.csv", "sha256": "67bf5d7f3bab8edd800d9acc2159c9c3e7cd01cbe72fddc0973ffc7e13f63f21", "size_bytes": 31744}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-e-ce-t050 / seed 2000: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-e-ce-t050/seed_2000/raw_samples.csv", "sha256": "252d36cfd849291b531924732e2a48f7e4bf2e375229797b76ff3a78b15d1095", "size_bytes": 32840}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-e-ce-t050 / seed 2001: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}, {"pid": 623773, "process_name": "/home/aidar.alimbayev/Documents/genmolv2/run_sources/running_mean_repair/.venv/bin/python", "used_memory_mib": 2154}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 3747, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-e-ce-t050/seed_2001/raw_samples.csv", "sha256": "423c7f380826ad0a36110e7f34cebc793981ff57a762f82c8c7f9d8db229589e", "size_bytes": 32663}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-mask-ce-t050 / seed 2000: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-mask-ce-t050/seed_2000/raw_samples.csv", "sha256": "2f7d7fc4d58888233de754024bc7d45cecb7ffa48d07747ac09c787ebcc884ce", "size_bytes": 33600}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

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